

GRADE 5 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 6) • Forward 2 (Grade 7)
71 question sets • 1930 questions

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Evaluate expressions that use parentheses

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting multi-digit numbers	4.NBT.B.4
FORWARD 1	Evaluate expressions with variables	6.EE.A.2.c
FORWARD 2	Evaluate expressions with negative numbers	7.NS.A.3

SET 1 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$6 \times (8 + 5)$$

Value: _____

Evaluate.

$$(35 - 11) \div 4$$

Value: _____

Evaluate.

$$50 - (6 \times 7)$$

Value: _____

Evaluate.

$$3 + (24 \div 6) \times 5$$

A. 35

B. 135

C. 27

D. 23

Evaluate.

$$(9 + 3) \times (10 - 4)$$

Value: _____

SET 1 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

True or false?

In $4 \times (2 + 5)$, you add $2 + 5$ first.

A. True

B. False

Evaluate.

$$(6 + 4) \times 2$$

Value: _____

Which step comes first in $18 - (3 \times 4)$?

- A.** Subtract $18 - 4$
- B.** Multiply 3×4
- C.** Subtract $18 - 3$
- D.** Add $3 + 4$

Evaluate.

$$20 - (5 + 5)$$

Value: _____

True or false?

$$(10 - 4) + 3 = 10 - 7$$

A. True

B. False

SET 1 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$7 \times 8$$

Product: _____

Divide.

$$24 \div 6$$

Quotient: _____

True or false?

$$9 \times 7 = 56$$

A. True

B. False

$$48 \div 8 = ?$$

A. 8

B. 5

C. 7

D. 6

Multiply.

$$3 \times 9$$

Product: _____

SET 1 • 5.OA.A.1

BACKWARD 3

Adding and subtracting multi-digit numbers • Grade 4 • Number & Operations
in Base Ten • 4.NBT.B.4

Add.

$$348 + 275$$

Sum: _____

Subtract.

$$502 - 168$$

Difference: _____

True or false?

$$1,250 - 375 = 985$$

A. True

B. False

$$609 + 194 = ?$$

A. 793

B. 703

C. 813

D. 803

Subtract.

$$1,000 - 432$$

Difference: _____

SET 1 • 5.OA.A.1

FORWARD 1

Evaluate expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4(x + 3)$ when $x = 5$.

Value: _____

Evaluate $20 - 2y$ when $y = 6$.

Value: _____

Evaluate $(a + b) \div 3$ when $a = 10$ and $b = 8$.

- A.** 4
- B.** 9
- C.** 54
- D.** 6

Evaluate $3m + 2$ when $m = 7$.

Value: _____

True or false?

When $n = 4$, the value of $5(n - 1)$ is 19.

A. True

B. False

SET 1 • 5.OA.A.1

FORWARD 2

Evaluate expressions with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$-3 \times (4 - 9)$$

Value: _____

Find the value.

$$(-12 + 4) \div 2$$

Value: _____

Find the value.

$$6 - (2 - 10)$$

A. -2

B. 2

C. 14

D. 18

Find the value.

$$(-5) \times (-2) + (-7)$$

Value: _____

True or false?

$$(-20) \div (3 + 2) = 4$$

A. True

B. False

Evaluate expressions with brackets and braces

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting within 100	2.NBT.B.5
FORWARD 1	Evaluate expressions with whole-number exponents	6.EE.A.1
FORWARD 2	Grouping symbols with negative numbers	7.NS.A.3

SET 2 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times [5 + (3 \times 4)]$$

Value: _____

Evaluate.

$$[20 - (2 + 6)] \div 3$$

Value: _____

Evaluate.

$$4 \times \{3 + [10 - (2 \times 4)]\}$$

A. 36

B. 20

C. 44

D. 26

Evaluate.

$$60 \div [3 \times (7 - 5)]$$

Value: _____

Evaluate.

$$\{18 - [2 \times (4 + 1)]\} + 9$$

Value: _____

SET 2 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(12 - 5) \times 3$$

Value: _____

True or false?

In $(8 + 2) \div 5$, you divide $2 \div 5$ first.

A. True

B. False

Evaluate.

$$4 \times (9 - 6)$$

A. 33

B. 10

C. 30

D. 12

Evaluate.

$$30 \div (2 + 4)$$

Value: _____

True or false?

$$(3 + 5) \times 2 = 16$$

A. True

B. False

SET 2 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$8 \times 6$$

Product: _____

Divide.

$$56 \div 7$$

Quotient: _____

True or false?

$$36 \div 4 = 8$$

A. True

B. False

$$7 \times 9 = ?$$

A. 63

B. 56

C. 72

D. 49

Divide.

$$81 \div 9$$

Quotient: _____

SET 2 • 5.OA.A.1

BACKWARD 3

Adding and subtracting within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$46 + 37$$

Sum: _____

Subtract.

$$72 - 28$$

Difference: _____

True or false?

$$55 + 29 = 74$$

A. True

B. False

$$90 - 47 = ?$$

A. 47

B. 43

C. 53

D. 33

Add.

$$38 + 54$$

Sum: _____

SET 2 • 5.OA.A.1

FORWARD 1

Evaluate expressions with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2 \times (3 + 1)^2$$

Value: _____

Evaluate.

$$[10 - 2^3] \times 5$$

Value: _____

Evaluate.

$$4^2 - (6 + 2)$$

- A.** 8
- B.** 2
- C.** 0
- D.** 24

Evaluate.

$$(5^2 - 1) \div 6$$

Value: _____

True or false?

$$3 \times 2^2 = 36$$

A. True

B. False

SET 2 • 5.OA.A.1

FORWARD 2

Grouping symbols with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$2 \times [-3 + (8 \div (-4))]$$

Value: _____

Find the value.

$$[(-6) + 15] \div (-3)$$

Value: _____

Find the value.

$$-4 \times [5 - (-1)]$$

A. 16

B. -16

C. -24

D. 24

Find the value.

$$\{10 - [2 \times (-3)]\} \div 4$$

Value: _____

True or false?

$$[-8 - (-2)] \times 3 = -30$$

A. True

B. False

Place grouping symbols to make an equation true

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	When regrouping keeps the same value	3.OA.B.5
BACKWARD 3	Multiplication and division facts	3.OA.C.7
FORWARD 1	Use parentheses to write equivalent expressions	6.EE.A.3
FORWARD 2	Expand and factor expressions with rational coefficients	7.EE.A.1

SET 3 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Insert parentheses to make the equation true.

$$4 + 6 \times 3 = 30$$

Equation: _____

Insert parentheses to make the equation true.

$$20 - 8 - 2 = 14$$

Equation: _____

Which placement of parentheses makes the equation true?

$$2 \times 5 + 3 \times 4 = 52$$

A. $2 \times (5 + 3) \times 4 = 52$

B. $(2 \times 5 + 3) \times 4 = 52$

C. $2 \times (5 + 3 \times 4) = 52$

D. $2 \times 5 + (3 \times 4) = 52$

Insert parentheses to make the equation true.

$$36 \div 6 - 2 = 9$$

Equation: _____

Insert brackets and parentheses to make the equation true.

$$2 \times 9 - 3 + 1 = 10$$

Equation: _____

SET 3 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(4 + 6) \times 3$$

Value: _____

Evaluate.

$$4 + (6 \times 3)$$

Value: _____

True or false?

$$(20 - 8) - 2 = 20 - (8 - 2)$$

A. True

B. False

Evaluate.

$$36 \div (6 - 2)$$

A. 4

B. 34

C. 9

D. 7

Evaluate.

$$(36 \div 6) - 2$$

Value: _____

SET 3 • 5.OA.A.1

BACKWARD 2

When regrouping keeps the same value • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

A. True

B. False

True or false?

$$(12 \div 6) \div 2 = 12 \div (6 \div 2)$$

A. True

B. False

Which equation is true?

A. $(5 + 2) + 3 = 5 + (2 + 3)$

B. $(9 - 4) - 1 = 9 - (4 - 1)$

C. $(16 \div 4) \div 2 = 16 \div (4 \div 2)$

D. $(7 - 3) - 2 = 7 - (3 - 2)$

True or false?

$$4 \times (5 \times 2) = (4 \times 5) \times 2$$

A. True

B. False

Fill in the blank.

$$(6 \times 2) \times 5 = 6 \times (2 \times \underline{\quad})$$

Blank: _____

SET 3 • 5.OA.A.1

BACKWARD 3

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$9 \times 4$$

Product: _____

Divide.

$$42 \div 6$$

Quotient: _____

True or false?

$$8 \times 7 = 54$$

A. True

B. False

$$64 \div 8 = ?$$

A. 6

B. 7

C. 8

D. 9

Multiply.

$$6 \times 7$$

Product: _____

SET 3 • 5.OA.A.1

FORWARD 1

Use parentheses to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(x + 5)$$

Expression: _____

Expand.

$$4(2y - 1)$$

Expression: _____

Which expression is equivalent to $6(a + 2)$?

- A.** $6a + 2$
- B.** $a + 12$
- C.** $6a + 12$
- D.** $8a$

Factor.

$$5m + 20$$

Expression: _____

True or false?

$$2(x + 7) = 2x + 7$$

A. True

B. False

SET 3 • 5.OA.A.1

FORWARD 2

Expand and factor expressions with rational coefficients • Grade 7 •
Expressions & Equations • 7.EE.A.1

Expand.

$$-2(3x - 4)$$

Expression: _____

Expand.

$$\frac{1}{2}(6y + 10)$$

Expression: _____

Which expression is equivalent to $-5(n + 2)$?

A. $-5n + 2$

B. $-5n - 10$

C. $-5n + 10$

D. $5n - 10$

Factor out 3.

$$3x - 12$$

Expression: _____

True or false?

$$0.5(4a - 8) = 2a - 8$$

A. True

B. False

Write numerical expressions from words

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Products as equal groups	3.OA.A.1
BACKWARD 2	Quotients as equal shares	3.OA.A.2
BACKWARD 3	Sums and differences	2.OA.A.1
FORWARD 1	Write expressions with variables	6.EE.A.2.a
FORWARD 2	Write and solve equations from words	7.EE.B.4.a

SET 4 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 9 and 6, then multiply by 4."

Expression: _____

Write an expression for "subtract 5 from 30, then divide by 5."

Expression: _____

Which expression shows "triple the sum of 12 and 8"?

A. $12 + 8 \times 3$

B. $3 \times 12 + 8$

C. $3 \times (12 + 8)$

D. $(3 + 12) \times 8$

Write an expression for "divide 40 by 8, then add 7."

Expression: _____

Write an expression for "the product of 6 and the difference of 15 and 9."

Expression: _____

SET 4 • 5.OA.A.2

BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

True or false?

5×7 can mean 5 groups of 7.

A. True

B. False

Which expression shows 4 groups of 6?

A. 46

B. $6 - 4$

C. $4 + 6$

D. 4×6

Write a multiplication expression for 8 groups of 3.

Expression: _____

True or false?

$$3 + 3 + 3 + 3 = 3 \times 3$$

A. True

B. False

What is the product of 7 and 5?

Product: _____

SET 4 • 5.OA.A.2

BACKWARD 2

Quotients as equal shares • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

True or false?

$15 \div 3$ can mean 15 shared equally into 3 groups.

A. True

B. False

Which expression shows 20 apples shared equally among 4 bags?

A. 20×4

B. $4 \div 20$

C. $20 \div 4$

D. $20 - 4$

What is the quotient of 18 and 6?

Quotient: _____

Write a division expression for 32 split into groups of 8.

Expression: _____

True or false?

The quotient of 12 and 2 is 24.

A. True

B. False

SET 4 • 5.OA.A.2

BACKWARD 3

Sums and differences • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

What is the sum of 14 and 9?

Sum: _____

What is the difference of 20 and 7?

Difference: _____

True or false?

The difference of 15 and 6 is 21.

A. True

B. False

Which word names the answer to an addition problem?

- A.** Difference
- B.** Quotient
- C.** Product
- D.** Sum

Write an expression for "7 more than 25."

Expression: _____

SET 4 • 5.OA.A.2

FORWARD 1

Write expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.a

Write an expression for "the sum of n and 8, times 3."

Expression: _____

Write an expression for "y divided by 5, then minus 2."

Expression: _____

Which expression means "4 less than the product of 2 and x"?

A. $4 - 2x$

B. $2x - 4$

C. $2(x - 4)$

D. $2 + x - 4$

Write an expression for "twice the difference of m and 6."

Expression: _____

True or false?

"9 more than k " can be written as $k + 9$.

A. True

B. False

SET 4 • 5.OA.A.2

FORWARD 2

Write and solve equations from words • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Three times the sum of a number and 4 is 27.

Write and solve an equation to find the number.

Equation and solution: _____

Half of a number, minus 6, is 10.

Write and solve an equation to find the number.

Equation and solution: _____

Which equation represents "2 less than 5 times a number is 33"?

A. $5n + 2 = 33$

B. $5n - 2 = 33$

C. $2 - 5n = 33$

D. $5(n - 2) = 33$

A number divided by 4, plus 3, is 9.

Write and solve an equation to find the number.

Equation and solution: _____

True or false?

The solution of $2(x + 3) = 18$ is $x = 7$.

A. True

B. False

Interpret numerical expressions without evaluating them

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Multiplicative comparison	4.OA.A.1
BACKWARD 2	Groups of a sum	3.OA.A.1
BACKWARD 3	Equal shares	3.OA.A.2
FORWARD 1	Identify parts of an expression	6.EE.A.2.b
FORWARD 2	Rewrite expressions to show how quantities are related	7.EE.A.2

SET 5 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

How does $4 \times (563 + 218)$ compare to $563 + 218$?

- A.** It is 4 more.
- B.** It is 4 times as large.
- C.** It is 4 less.
- D.** It is one-fourth as large.

True or false?

$(7,215 - 1,005) \div 5$ is one-fifth as large as $7,215 - 1,005$.

A. True

B. False

Without calculating, how many times as large is $9 \times (48 + 27)$ as $48 + 27$?

Times as large: _____

Which expression is twice as large as 356×12 ?

A. $(356 + 2) \times 12$

B. $356 \times 12 + 2$

C. 356×14

D. $2 \times (356 \times 12)$

True or false?

$6 \times 25 + 4$ is 6 times as large as $25 + 4$.

A. True

B. False

SET 5 • 5.OA.A.2

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.1

True or false?

35 is 5 times as many as 7.

A. True

B. False

Fill in the blank.

48 is ____ times as many as 6.

Blank: _____

Which equation shows "36 is 4 times as many as 9"?

A. $9 = 4 \times 36$

B. $36 = 4 + 9$

C. $36 - 9 = 4$

D. $36 = 4 \times 9$

True or false?

30 is 3 times as many as 27.

A. True

B. False

Tom has 6 cards. Ana has 3 times as many cards.
How many cards does Ana have?

Cards: _____

SET 5 • 5.OA.A.2

BACKWARD 2

Groups of a sum • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 1)$ means 3 groups of $(4 + 1)$.

A. True

B. False

Which expression means 6 groups of $(10 + 2)$?

A. $6 \times (10 + 2)$

B. $(6 + 10) \times 2$

C. $6 \times 10 + 2$

D. $6 + (10 + 2)$

How many groups of $(5 + 4)$ are in $7 \times (5 + 4)$?

Groups: _____

True or false?

$$2 \times (8 + 1) = (8 + 1) + 2$$

A. True

B. False

Write 4 groups of 9 as a multiplication expression.

Expression: _____

SET 5 • 5.OA.A.2

BACKWARD 3

Equal shares • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

True or false?

12 shared equally among 3 people is 4 each.

A. True

B. False

A 20-inch ribbon is cut into 5 equal pieces.
How long is each piece?

Length: _____

Which expression finds one of 4 equal shares of 28?

A. $28 - 4$

B. $28 \div 4$

C. 28×4

D. $4 \div 28$

True or false?

Dividing a number by 2 splits it into 2 equal shares.

A. True

B. False

30 crayons are shared equally among 6 children.

How many crayons does each child get?

Crayons: _____

SET 5 • 5.OA.A.2

FORWARD 1

Identify parts of an expression • Grade 6 • Expressions & Equations •
6.EE.A.2.b

In $5(x + 2)$, what is 5?

- A.** A term of $x + 2$
- B.** A quotient
- C.** A sum
- D.** A factor

How many terms are in $4a + 7b - 1$?

Terms: _____

What is the coefficient of m in $9m + 4$?

Coefficient: _____

True or false?

In $3(y + 6)$, the quantity $(y + 6)$ is a sum of two terms.

A. True

B. False

What is the expression $12 \div (n + 1)$ called?

- A.** A coefficient
- B.** A quotient
- C.** A sum
- D.** A product

SET 5 • 5.OA.A.2

FORWARD 2

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price p is increased by 5%.

Which expression shows the new price?

A. $0.05p$

B. $1.05p$

C. $5p$

D. $p + 5$

True or false?

$$x + 0.3x = 0.4x$$

A. True

B. False

Write $n - 0.4n$ as one term.

Expression: _____

Which expression is equivalent to $2(3t)$?

- A.** $5t$
- B.** $2t + 3t$
- C.** $6t$
- D.** $3t + 2$

A sale price is $0.75p$, where p is the original price.
What percent of the original price is taken off?

Percent: _____

Generate two numerical patterns and compare their terms

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Equations relating dependent and independent variables	6.EE.C.9
FORWARD 2	Equations for proportional relationships	7.RP.A.2.c

SET 6 • 5.OA.B.3

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 6.

Pattern B starts at 0 and adds 18.

Write the first 5 terms of each pattern.

Pattern A: _____

Pattern B: _____

Pattern A: 0, 4, 8, 12, 16

Pattern B: 0, 12, 24, 36, 48

How does each term in Pattern B compare to the matching term in Pattern A?

A. 8 more

B. 12 more

C. 4 times as large

D. 3 times as large

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10.

What is the 6th term of each pattern?

Pattern A: _____

Pattern B: _____

True or false?

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8. Each term in Pattern B is 4 times the matching term in Pattern A.

- A.** True
- B.** False

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

When Pattern A is 28, what is the matching term in Pattern B?

Term: _____

SET 6 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 6." The pattern starts at 0.
Write the first five terms.

Terms: _____

True or false?

The rule "add 4," starting at 0, gives 0, 4, 8, 16.

A. True

B. False

What is the rule for the pattern?

0, 9, 18, 27, 36

- A.** Add 9
- B.** Add 18
- C.** Multiply by 9
- D.** Add 8

The rule is "add 11." The pattern starts at 0.
What is the 4th term?

Term: _____

What comes next?

0, 15, 30, 45, ____

Next term: _____

SET 6 • 5.OA.B.3

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$4 \times 7$$

Product: _____

True or false?

$$6 \times 8 = 42$$

A. True

B. False

What number times 3 equals 24?

- A.** 8
- B.** 7
- C.** 21
- D.** 6

Multiply.

$$5 \times 9$$

Product: _____

What number times 6 equals 42?

Number: _____

SET 6 • 5.OA.B.3

FORWARD 1

Equations relating dependent and independent variables • Grade 6 •
Expressions & Equations • 6.EE.C.9

Each box holds 6 cans.

Write an equation for the number of cans c in b boxes.

Equation: _____

In $c = 6b$, which variable is the dependent variable?

- A.** b
- B.** c
- C.** 6
- D.** Neither

Write an equation for y in terms of x .

x	1	2	3	4
y	5	10	15	20

Equation: _____

True or false?

In $y = x + 4$, y is the independent variable.

A. True

B. False

A plant grows 2 cm each week.

Write an equation for its growth g after w weeks.

Equation: _____

SET 6 • 5.OA.B.3

FORWARD 2

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost \$0.35 each.

Write an equation for the cost c of n pencils.

Equation: _____

The table shows a proportional relationship.

Write an equation for y in terms of x .

x	4	6	10
y	10	15	25

Equation: _____

Which equation represents a proportional relationship?

A. $y = 1.5$

B. $y = x + 1.5$

C. $y = 1.5x + 2$

D. $y = 1.5x$

A train travels 150 miles in 2 hours at a constant speed.
Write an equation for the distance d after t hours.

Equation: _____

True or false?

$y = \frac{3}{4}x$ represents a proportional relationship.

A. True

B. False

Explain why two patterns have the relationship they do

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Notice features of a pattern that the rule does not state	4.OA.C.5
BACKWARD 2	Explain patterns in the multiplication table	3.OA.D.9
FORWARD 1	Use an equation to describe how two quantities change together	6.EE.C.9
FORWARD 2	Identify the constant of proportionality	7.RP.A.2.b

SET 58 • 5.OA.B.3

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8.

Each term in Pattern B is twice the matching term in Pattern A.

Explain why.

Explanation: _____

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 15.

Why is each term in Pattern B 5 times the matching term in Pattern A?

- A.** Both start at 0, and B adds 5 times as much as A at every step.
- B.** Pattern B starts 5 higher than Pattern A.
- C.** Pattern B has 5 more terms than Pattern A.
- D.** Each term in B is 12 more than the matching term in A.

True or false?

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6. Each term in B is 3 times the matching term in A because B adds 3 times as much at every step.

A. True

B. False

True or false?

Pattern A starts at 1 and adds 2. Pattern B starts at 1 and adds 4. Each term in Pattern B is twice the matching term in Pattern A.

A. True

B. False

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 20.

Without listing the terms, how does the 10th term of B compare with the 10th term of A? Explain.

Comparison: _____

Explanation: _____

SET 58 • 5.OA.B.3

BACKWARD 1

Notice features of a pattern that the rule does not state • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

True or false?

The rule is "add 2," starting at 1. Every term is odd.

A. True

B. False

The rule is "add 4," starting at 0. Which is true of every term?

- A.** It is a multiple of 4.
- B.** It is odd.
- C.** It ends in 4.
- D.** It is a multiple of 8.

The rule is "add 10," starting at 5.

What do you notice about the ones digit of every term?

Answer: _____

True or false?

The rule is "add 5," starting at 0. The terms alternate between ending in 0 and ending in 5.

A. True

B. False

The rule is "add 3," starting at 1: 1, 4, 7, 10, 13.

Which describes the terms?

- A.** They alternate odd, even, odd, even.
- B.** They are all odd.
- C.** They are all even.
- D.** They are all multiples of 3.

SET 58 • 5.OA.B.3

BACKWARD 2

Explain patterns in the multiplication table • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

True or false?

Every product in the 4s row of the multiplication table is even.

- A. True
- B. False

Why is each product in the 6s row twice the product in the 3s row, in the same column?

- A.** 6 is 2×3 , so $6 \times n$ is $2 \times (3 \times n)$.
- B.** 6 is 3 more than 3.
- C.** The 6s row has twice as many products.
- D.** Products in later rows are always even.

True or false?

Every product in the 5s row ends in 5.

A. True

B. False

The 2s row is 2, 4, 6, 8, 10, ...

What is the same about every product?

Answer: _____

True or false?

In the 9s row, 9×4 is 9 more than 9×3 .

A. True

B. False

SET 58 • 5.OA.B.3

FORWARD 1

Use an equation to describe how two quantities change together • Grade 6 •
Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x .

Explain how the table shows it.

x	1	2	3	4
y	7	14	21	28

Equation: _____

Explanation: _____

Each time x increases by 1, y increases by 5, and y is 0 when x is 0.

Which equation fits?

A. $y = 5x$

B. $y = 5$

C. $y = x + 5$

D. $x = 5y$

True or false?

In $y = x + 3$, each y -value is 3 times its x -value.

A. True

B. False

A gym charges \$9 for each class.

Write an equation for the cost c of n classes.

Equation: _____

True or false?

For $y = 4x$, when x doubles, y also doubles.

- A. True
- B. False

SET 58 • 5.OA.B.3

FORWARD 2

Identify the constant of proportionality • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality?

x	2	5	8
y	6	15	24

Constant: _____

What is the constant of proportionality in $y = 0.6x$?

- A.** 0.6
- B.** 0
- C.** 6
- D.** 1.6

True or false?

In the table x : 4, 6, 10 and y : 2, 3, 5, the constant of proportionality is 2.

- A. True
- B. False

A car travels 50 miles on every 2 gallons of gas.
What is the constant of proportionality, in miles per gallon?

Constant: _____

True or false?

A proportional relationship graphs as a line through $(0, 0)$ and $(1, 7)$. Its constant of proportionality is 7.

- A.** True
- B.** False

Form ordered pairs from two patterns and graph them

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 3	Read a scaled bar graph	3.MD.B.3
BACKWARD 4	Whole numbers on a number line	2.MD.B.6
FORWARD 1	Plot pairs of values from a ratio table	6.RP.A.3.a
FORWARD 2	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a

SET 7 • 5.OA.B.3

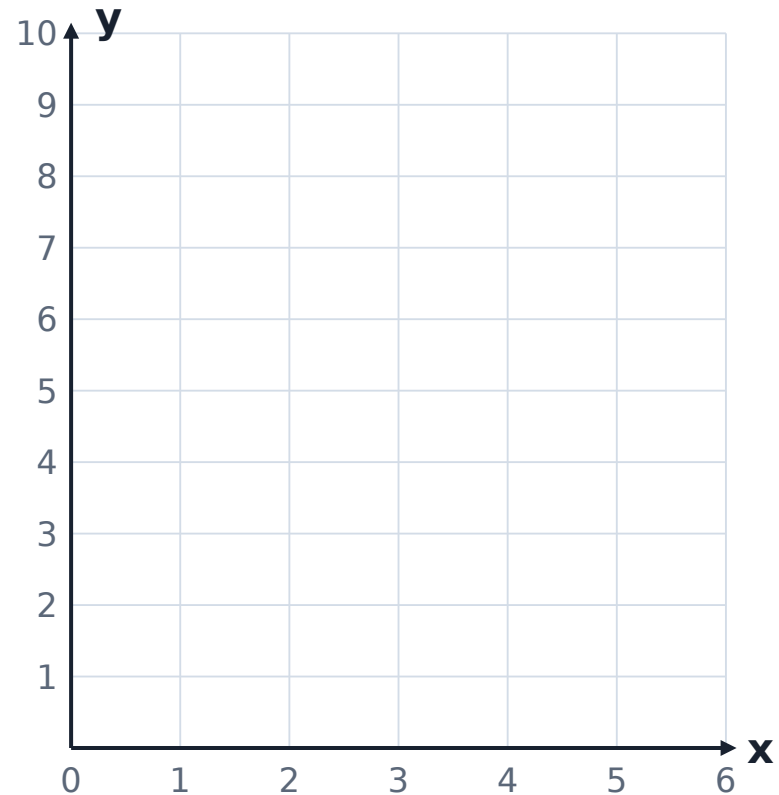
MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A (x): start at 0 and add 1.

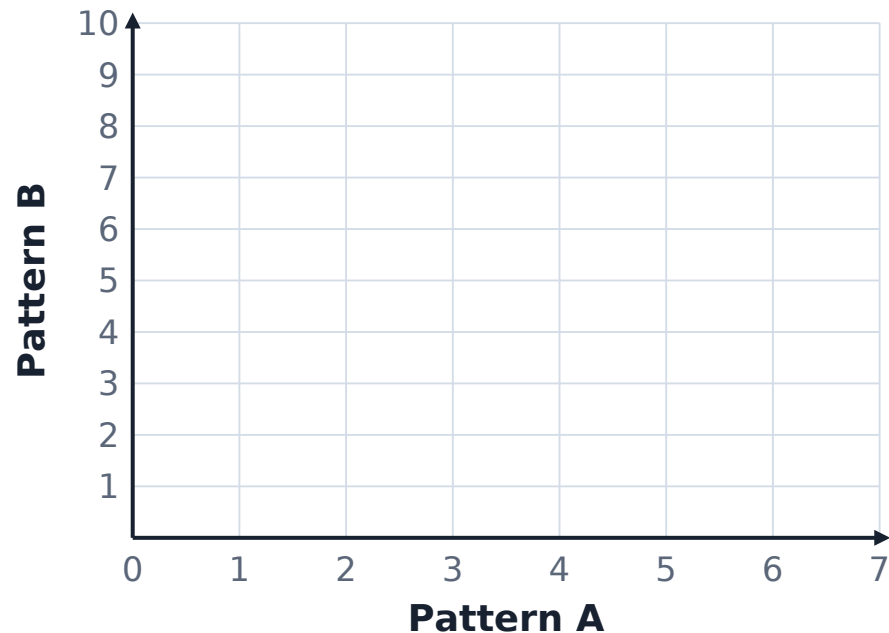
Pattern B (y): start at 0 and add 2.

Graph the ordered pairs for the first 5 terms.



Graph the ordered pairs from the two patterns.

Pattern A	0	2	4	6
Pattern B	0	3	6	9



Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

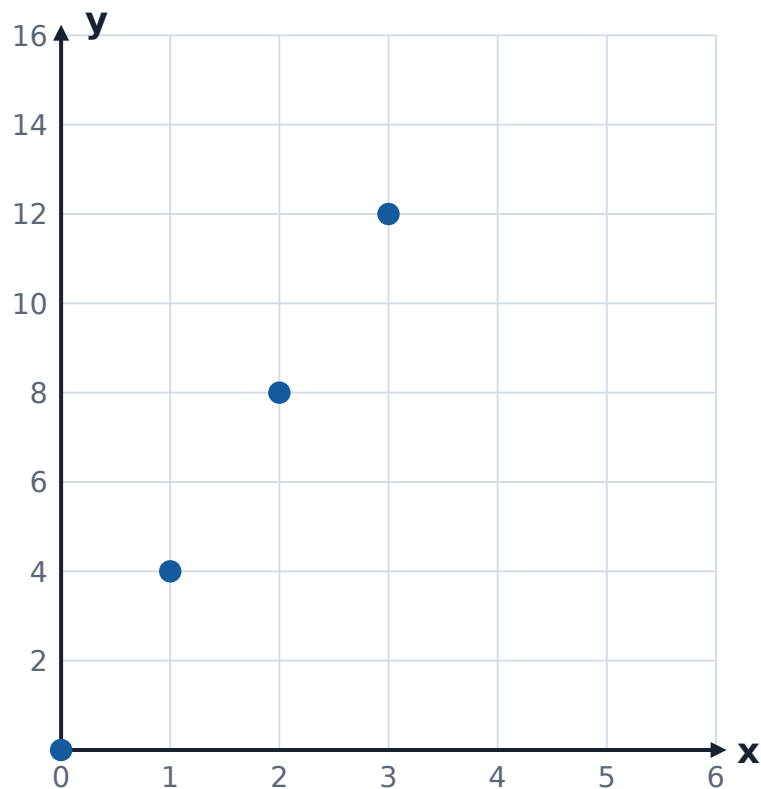
Write the first four ordered pairs (A, B).

Ordered pairs: _____

The points come from two patterns.

Which statement is true?

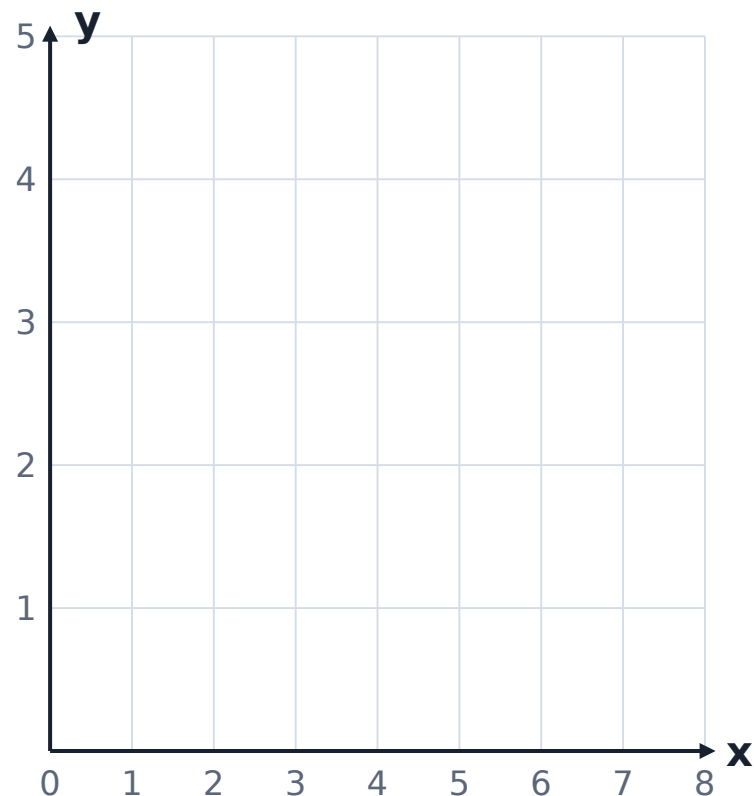
- A.** The y-values increase by 1 each time.
- B.** Each y-value is 4 times its x-value.
- C.** Each x-value is 4 times its y-value.
- D.** Each y-value is 4 more than its x-value.



Pattern A (x): start at 0 and add 2.

Pattern B (y): start at 0 and add 1.

Graph the ordered pairs for the first 4 terms.



SET 7 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 3." The pattern starts at 0.
What is the 5th term?

Term: _____

True or false?

The pattern 0, 5, 10, 15 follows the rule "add 5."

A. True

B. False

Which pattern follows the rule "add 2," starting at 0?

A. 0, 2, 4, 6

B. 2, 4, 6, 8

C. 0, 1, 2, 3

D. 0, 2, 4, 8

What is the rule for the pattern?

0, 8, 16, 24

Rule: _____

The rule is "add 10." The pattern starts at 0.
What is the 6th term?

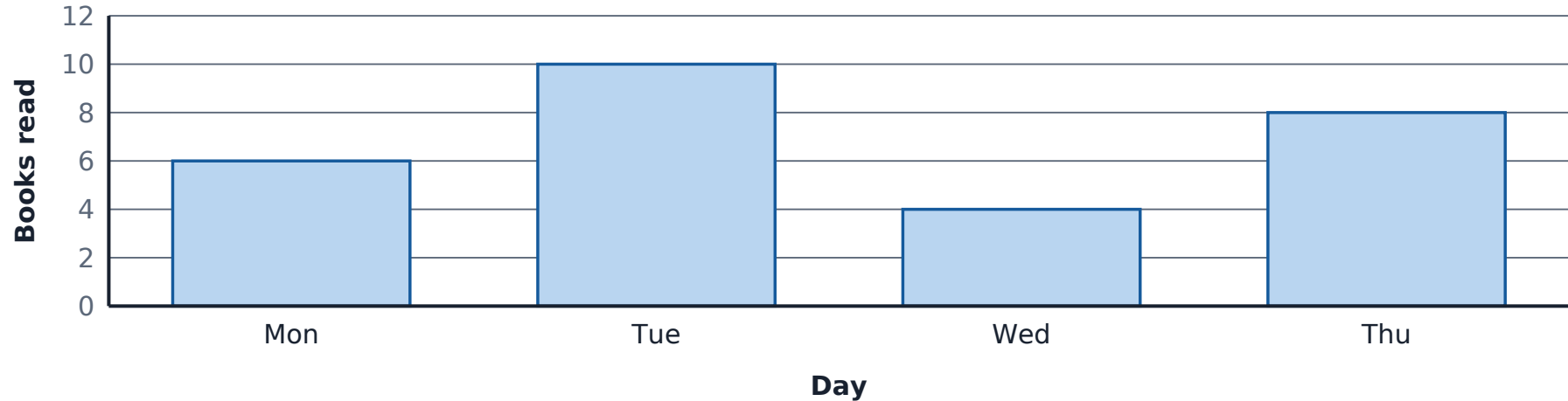
Term: _____

SET 7 • 5.OA.B.3

BACKWARD 3

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many books does each grid line on the scale stand for?



A. 4

B. 1

C. 12

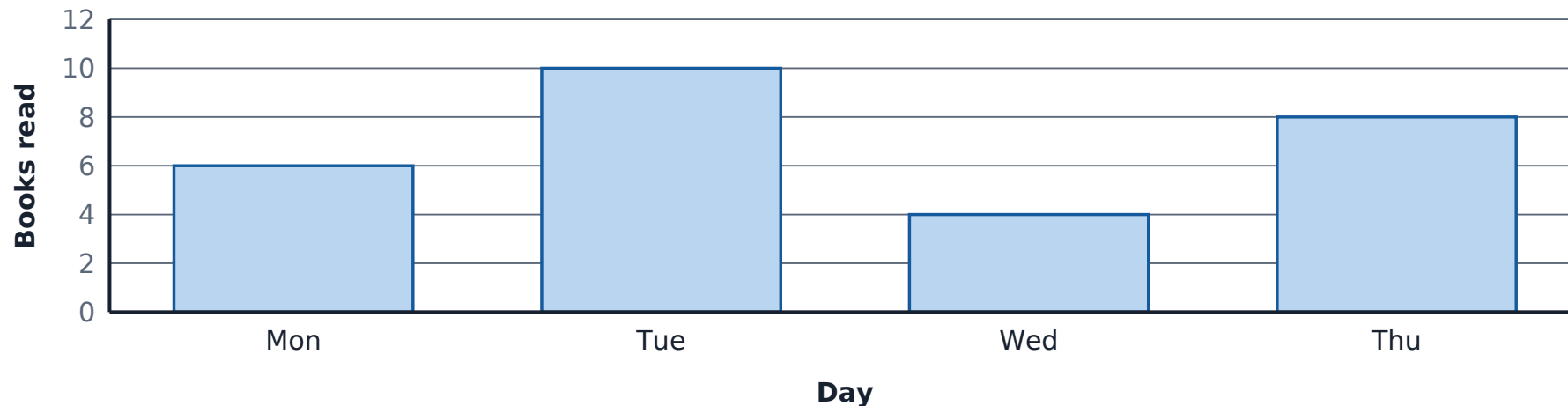
D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

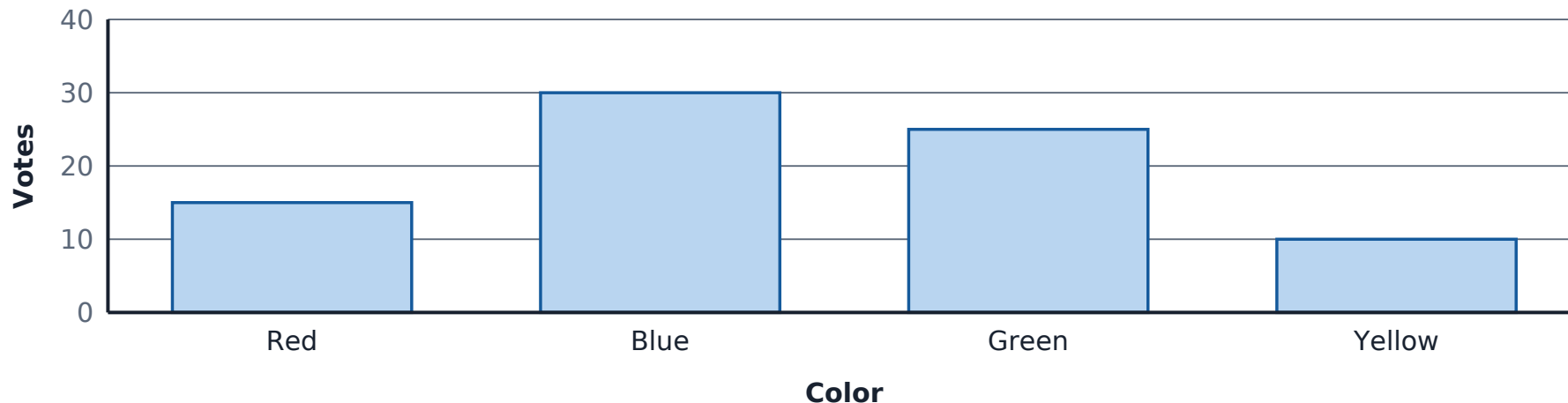
Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q1/5

How many books were read on Tuesday?



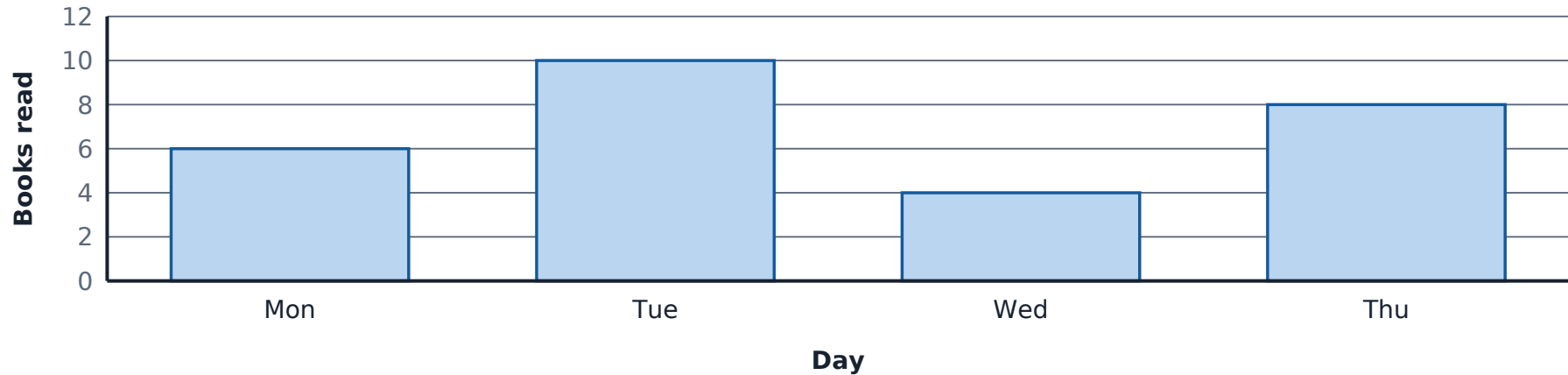
Books: _____

How many votes did Green get?



Votes: _____

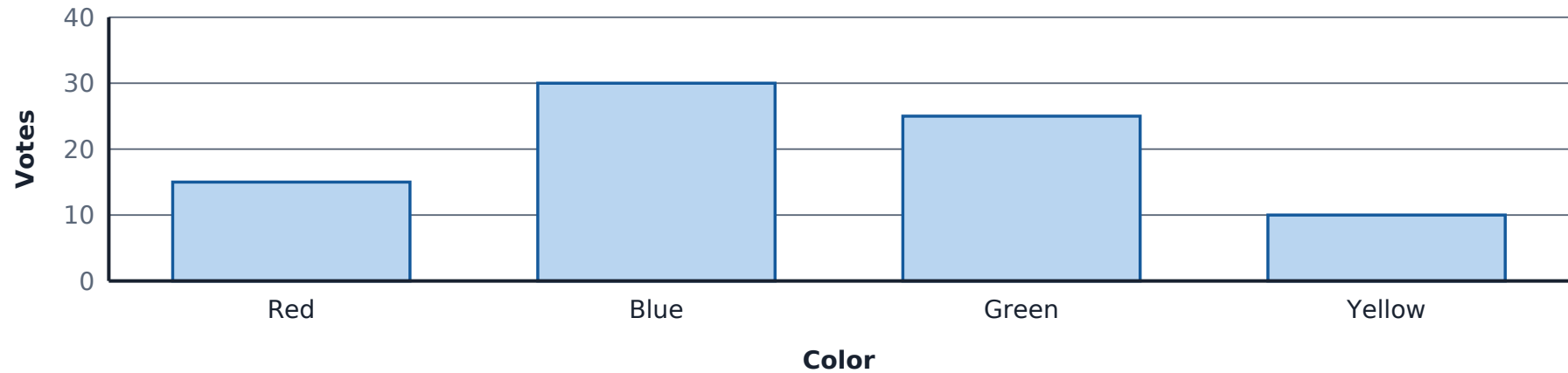
How many more books were read on Tuesday than on Wednesday?



Books: _____

True or false?

Red got 5 more votes than Yellow.



A. True

B. False

SET 7 • 5.OA.B.3

BACKWARD 4

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

True or false?

From 0 to 4 on this number line there are 4 equal spaces.



A. True

B. False

The spaces on a number line are equal and each stands for 1.

Which number is 3 spaces to the right of 5?

A. 8

B. 2

C. 53

D. 15

What number is at point B?



B = _____

True or false?

On a number line, 9 is to the left of 6.

A. True

B. False

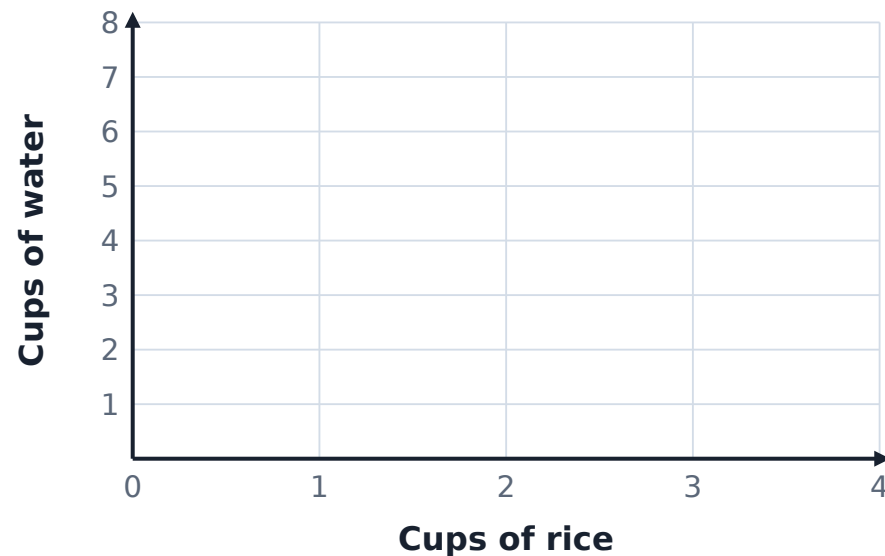
SET 7 • 5.OA.B.3

FORWARD 1

Plot pairs of values from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Plot the pairs of values from the ratio table on the coordinate plane.

Cups of rice	1	2	3
Cups of water	2	4	6



The table shows equivalent ratios.

Find the missing value.

Laps	2	4	6	8
Minutes	5	10	?	20

Missing value: _____

Which point comes from the ratio table?

Boxes	1	2	3
Pens	6	12	18

A. (2, 6)

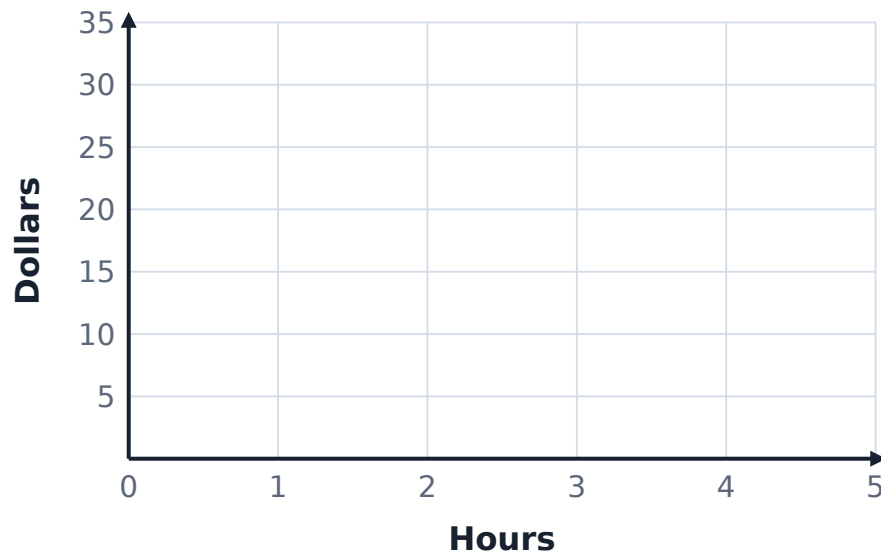
B. (12, 2)

C. (2, 12)

D. (3, 12)

Plot the pairs of values from the ratio table on the coordinate plane.

Hours	1	2	3	4
Dollars	8	16	24	32



True or false?

The points (1, 3), (2, 6), and (3, 9) come from a ratio table with the ratio 1 : 3.

A. True

B. False

SET 7 • 5.OA.B.3

FORWARD 2

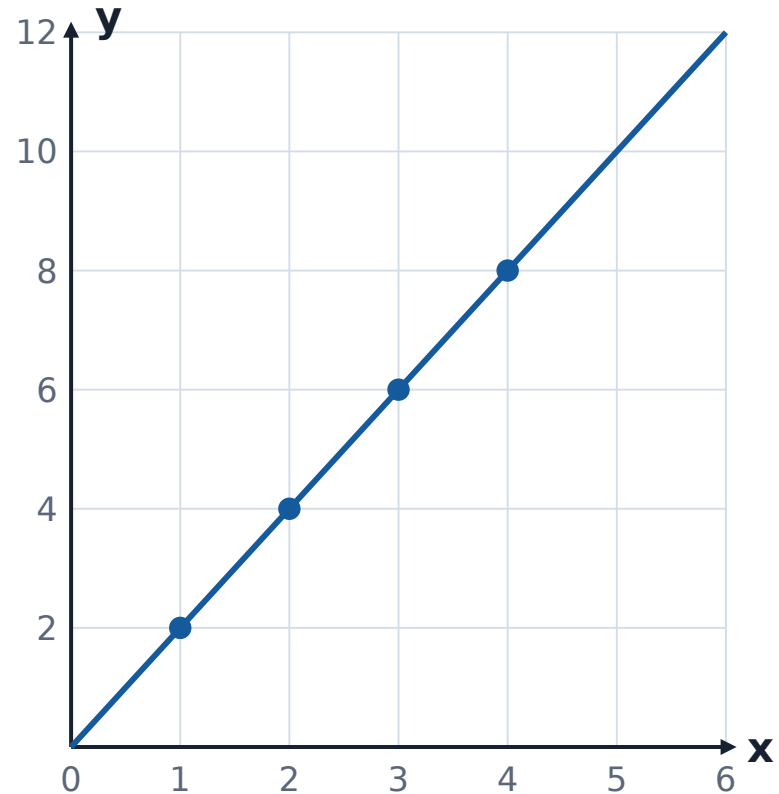
Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The points lie on a straight line through the origin, so the relationship is proportional.

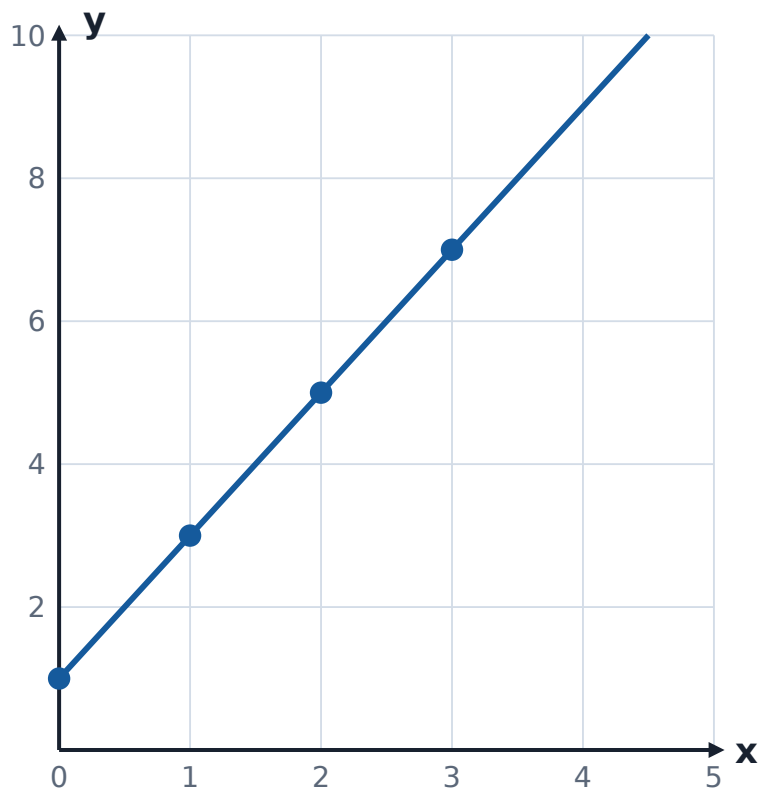
A. True

B. False



Which statement about the graph is true?

- A.** It is not proportional because the line does not pass through $(0, 0)$.
- B.** It is not proportional because the points are not on a line.
- C.** It is proportional because the points are on a line.
- D.** It is proportional because y increases by 2.

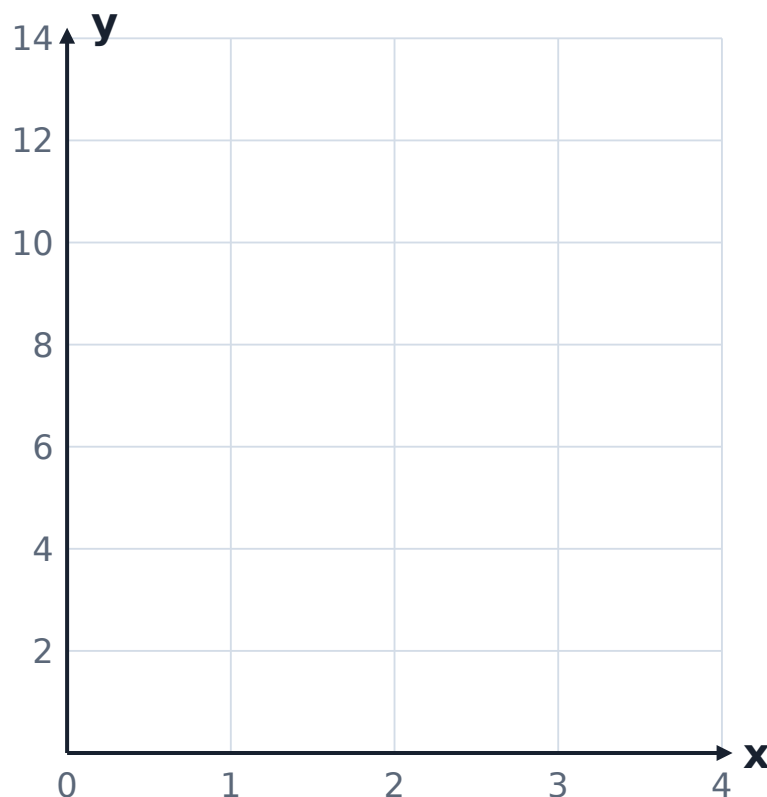


Graph the pairs $(1, 4)$, $(2, 8)$, and $(3, 12)$.

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____



True or false?

The points (1, 2), (2, 5), and (3, 8) show a proportional relationship.

- A. True
- B. False

Which set of points shows a proportional relationship?

- A.** (1, 1), (2, 4), (3, 9)
- B.** (1, 2), (2, 3), (3, 4)
- C.** (0, 3), (1, 6), (2, 9)
- D.** (2, 5), (4, 10), (6, 15)

A digit is worth 10 times as much as the same digit one place to its right

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	Each place is worth 10 times the place to its right	4.NBT.A.1
BACKWARD 2	Tenths and hundredths as decimals	4.NF.C.6
BACKWARD 3	Tenths as hundredths	4.NF.C.5
FORWARD 1	Compute with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 8 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

In 5,550, how does the value of the 5 in the hundreds place compare to the value of the 5 in the tens place?

- A.** It is 100 times as much.
- B.** It is 10 times as much.
- C.** It is the same.
- D.** It is $\frac{1}{10}$ as much.

In 3.33, the first 3 is in the ones place and the second 3 is in the tenths place.

How many times as much is the first 3 worth as the second 3?

Times as much: _____

True or false?

In 0.77, the 7 in the tenths place is worth 10 times as much as the 7 in the hundredths place.

- A. True
- B. False

Find the value of the 4 in 642.18 and the value of the 4 in 6,421.8.

How many times as much is the second 4 worth?

4 in 642.18: _____

4 in 6,421.8: _____

Times as much: _____

In which number is the 6 worth 10 times as much as the 6 in 3.6?

A. 0.36

B. 3.06

C. 46.1

D. 60.3

SET 8 • 5.NBT.A.1

BACKWARD 1

Each place is worth 10 times the place to its right • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

In 880, the 8 in the hundreds place is worth 10 times as much as the 8 in the tens place.

A. True

B. False

What is the value of the 7 in 2,740?

Value: _____

Which number is 10 times as much as 60?

- A.** 600
- B.** 6,000
- C.** 70
- D.** 6

How many tens are in 1 hundred?

Tens: _____

True or false?

70 is 10 times as much as 7.

A. True

B. False

SET 8 • 5.NBT.A.1

BACKWARD 2

Tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{3}{10}$ as a decimal.

Decimal: _____

Write $\frac{47}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.6 = \frac{6}{100}$$

A. True

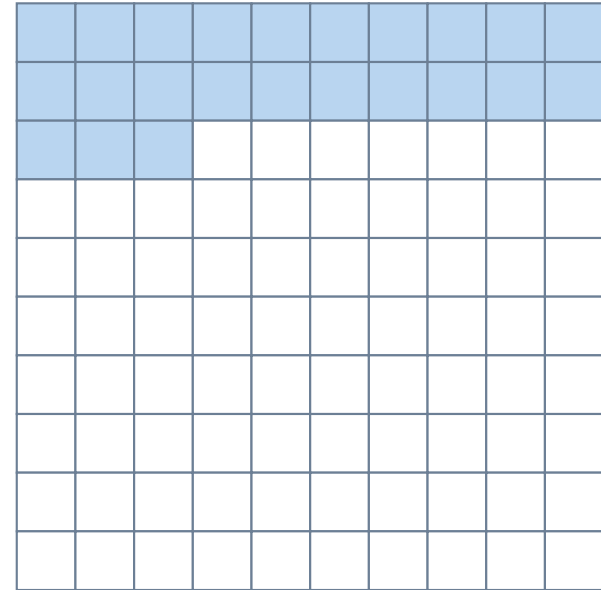
B. False

Which decimal shows 5 hundredths?

- A.** 0.5
- B.** 0.05
- C.** 0.005
- D.** 5.0

What decimal does the shaded part of the grid show?

Decimal: _____



SET 8 • 5.NBT.A.1

BACKWARD 3

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

True or false?

$$\frac{1}{10} = \frac{10}{100}$$

A. True

B. False

How many hundredths are in 1 tenth?

Hundredths: _____

Which fraction is equal to $\frac{4}{10}$?

A. $\frac{4}{100}$

B. $\frac{400}{100}$

C. $\frac{14}{100}$

D. $\frac{40}{100}$

Find the missing number.

$$\frac{7}{10} = \frac{?}{100}$$

Missing number: _____

True or false?

$$\frac{3}{10} + \frac{5}{100} = \frac{35}{100}$$

A. True

B. False

SET 8 • 5.NBT.A.1

FORWARD 1

Compute with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

$$27.6 + 4.38$$

Sum: _____

Subtract.

$$50.2 - 7.65$$

Difference: _____

Multiply.

$$1.2 \times 0.4$$

A. 4.8

B. 0.48

C. 0.048

D. 1.6

Divide.

$$7.5 \div 2.5$$

Quotient: _____

True or false?

$$0.3 \times 0.3 = 0.9$$

A. True

B. False

SET 8 • 5.NBT.A.1

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver at -4.5 m descends 2.25 m.

What is the diver's new position?

Position: _____

The temperature was 3.5°F . It fell 6.2°F .
What is the new temperature?

Temperature: _____

A stock's value changed by $-\$0.75$ each day for 4 days.
What was the total change?

- A.** $-\$0.30$
- B.** $-\$4.75$
- C.** $\$3.00$
- D.** $-\$3.00$

A bank balance of $-\$12.40$ increases by $\$20.00$.

What is the new balance?

Balance: _____

True or false?

$$-1.5 \times 4 = 6$$

A. True

B. False

A digit is worth $\frac{1}{10}$ of the same digit one place to its left

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	One place to the right is worth $\frac{1}{10}$ as much	4.NBT.A.1
BACKWARD 2	Unit fractions	3.NF.A.1
BACKWARD 3	Decimals for tenths and hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Write fractions as decimals using long division	7.NS.A.2.d

SET 9 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

True or false?

In 44.4, the 4 in the tenths place is worth $\frac{1}{10}$ of the 4 in the ones place.

- A. True
- B. False

In 2.22, the 2 in the hundredths place is worth what part of the 2 in the tenths place?

A. 10 times as much

B. $\frac{1}{10}$

C. The same amount

D. $\frac{1}{100}$

What is $\frac{1}{10}$ of 50?

Answer: _____

What is $\frac{1}{10}$ of 0.8?

Answer: _____

In which number is the 9 worth $\frac{1}{10}$ of the 9 in 19.4?

- A.** 5.92
- B.** 3.009
- C.** 9.01
- D.** 90.1

SET 9 • 5.NBT.A.1

BACKWARD 1

One place to the right is worth $\frac{1}{10}$ as much • Grade 4 • Number & Operations
in Base Ten • 4.NBT.A.1

True or false?

In 330, the 3 in the tens place is worth $\frac{1}{10}$ of the 3 in the hundreds place.

- A.** True
- B.** False

Divide.

$$400 \div 10$$

Quotient: _____

Which number is $\frac{1}{10}$ of 3,000?

- A.** 3
- B.** 300
- C.** 30,000
- D.** 30

How many hundreds are in 1 thousand?

Hundreds: _____

True or false?

$$60 \div 10 = 600$$

A. True

B. False

SET 9 • 5.NBT.A.1

BACKWARD 2

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{8}$ means 1 of 8 equal parts of a whole.

A. True

B. False

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

Which fraction is greater?

A. $\frac{1}{8}$

B. $\frac{1}{2}$

True or false?

A whole cut into 4 equal parts has parts that are each $\frac{1}{3}$ of the whole.

- A. True
- B. False

What fraction of the bar is shaded?



Fraction: _____

SET 9 • 5.NBT.A.1

BACKWARD 3

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{9}{100}$ as a decimal.

Decimal: _____

Write 0.8 as a fraction.

Fraction: _____

True or false?

0.4 and 0.04 have the same value.

A. True

B. False

Which decimal is equal to $\frac{6}{10}$?

A. 6.0

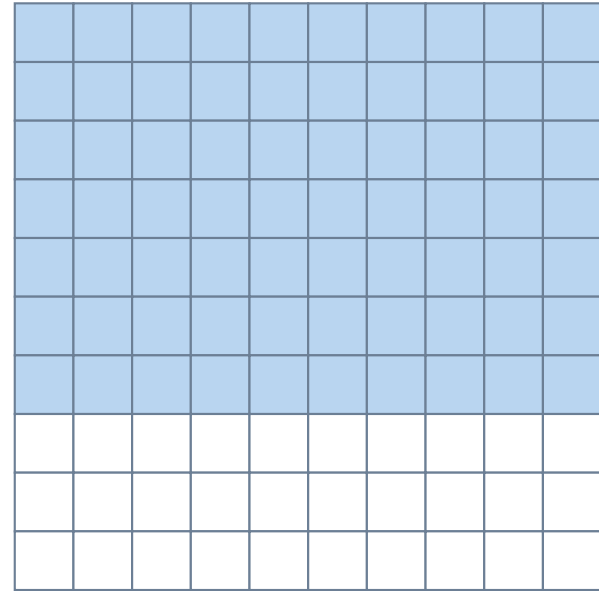
B. 0.06

C. 6.10

D. 0.6

What decimal does the shaded part of the grid show?

Decimal: _____



SET 9 • 5.NBT.A.1

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$4.8 \div 0.4$$

Quotient: _____

Divide.

$$9.36 \div 3$$

Quotient: _____

Divide.

$$0.56 \div 0.7$$

A. 80

B. 0.08

C. 8

D. 0.8

Divide.

$$15 \div 0.25$$

Quotient: _____

True or false?

$$6.3 \div 0.9 = 0.7$$

A. True

B. False

SET 9 • 5.NBT.A.1

FORWARD 2

Write fractions as decimals using long division • Grade 7 • The Number System
• 7.NS.A.2.d

Use long division to write $\frac{7}{8}$ as a decimal. Show your work.

Decimal: _____

Use long division to write $\frac{2}{9}$ as a decimal. Show your work.

Decimal: _____

Which fraction has a terminating decimal?

A. $\frac{5}{6}$

B. $\frac{4}{7}$

C. $\frac{1}{3}$

D. $\frac{3}{20}$

Use long division to write $\frac{9}{20}$ as a decimal. Show your work.

Decimal: _____

True or false?

The decimal form of $\frac{1}{11}$ repeats.

A. True

B. False

Multiply and divide whole numbers by powers of 10; write powers of 10 with exponents

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Multiplying and dividing by 10	4.NBT.A.1
BACKWARD 2	Multiplying by multiples of 10	3.NBT.A.3
FORWARD 1	Write and evaluate whole-number exponents	6.EE.A.1
FORWARD 2	Multiply and divide negative numbers by powers of 10 — nearest...	7.NS.A.2.c

SET 10 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Multiply.

$$37 \times 1,000$$

Product: _____

Divide.

$$45,000 \div 100$$

Quotient: _____

Write $10 \times 10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

$$6 \times 10^3 = ?$$

A. 60,000

B. 6,000

C. 18

D. 600

True or false?

$$820 \div 10^2 = 82$$

A. True

B. False

SET 10 • 5.NBT.A.2

BACKWARD 1

Multiplying and dividing by 10 • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

Multiply.

$$25 \times 10$$

Product: _____

True or false?

300 is 10 times as much as 30.

A. True

B. False

Which number is 10 times as much as 450?

- A.** 4,500
- B.** 45,000
- C.** 460
- D.** 45

How many tens are in 800?

Tens: _____

Divide.

$$7,000 \div 10$$

Quotient: _____

SET 10 • 5.NBT.A.2

BACKWARD 2

Multiplying by multiples of 10 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.3

Multiply.

$$4 \times 60$$

Product: _____

Multiply.

$$9 \times 80$$

Product: _____

True or false?

$$7 \times 30 = 2,100$$

A. True

B. False

$$5 \times 90 = ?$$

A. 4,500

B. 450

C. 45

D. 95

Multiply.

$$8 \times 70$$

Product: _____

SET 10 • 5.NBT.A.2

FORWARD 1

Write and evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$3^4$$

Value: _____

Write $5 \times 5 \times 5$ using an exponent.

Exponent form: _____

$$2^5 = ?$$

A. 25

B. 10

C. 64

D. 32

Evaluate.

$$10^3 + 2^2$$

Value: _____

True or false?

$$4^3 = 12$$

A. True

B. False

FORWARD 2

Multiply and divide negative numbers by powers of 10 • Grade 7 • The Number System • 7.NS.A.2.c

▮ No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Multiply.

$$-3.5 \times 100$$

Product: _____

Divide.

$$-420 \div 10$$

Quotient: _____

$$-0.06 \times 1,000 = ?$$

- A.** -60
- B.** -0.006
- C.** -6
- D.** 60

Divide.

$$7.2 \div (-10)$$

Quotient: _____

True or false?

$$(-10) \times (-10) \times (-10) = -1,000$$

A. True

B. False

Place the decimal point when multiplying or dividing a decimal by a power of 10

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Decimal notation for fractions	4.NF.C.6
BACKWARD 2	Place value when multiplying by 10 or 100	4.NBT.A.1
FORWARD 1	Compute fluently with multi-digit decimals	6.NS.B.3
FORWARD 2	Powers of 10 with negative rational numbers — nearest related (n...	7.NS.A.2.c

SET 11 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Multiply.

$$3.82 \times 100$$

Product: _____

Divide.

$$56.4 \div 10$$

Quotient: _____

Multiply.

$$0.075 \times 10^2$$

Product: _____

Divide.

$$2.9 \div 10^3$$

A. 0.029

B. 0.29

C. 0.0029

D. 2,900

True or false?

When you multiply a number by 1,000, each digit moves 3 places to the left.

A. True

B. False

SET 11 • 5.NBT.A.2

BACKWARD 1

Decimal notation for fractions • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

Write $\frac{35}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.7 = \frac{7}{100}$$

A. True

B. False

Which decimal is $2\frac{4}{10}$?

A. 24

B. 2.4

C. 0.24

D. 2.04

Write 0.09 as a fraction.

Fraction: _____

True or false?

$$1.5 = 1\frac{5}{100}$$

A. True

B. False

SET 11 • 5.NBT.A.2

BACKWARD 2

Place value when multiplying by 10 or 100 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

When a whole number is multiplied by 10, each digit moves one place to the left.

A. True

B. False

Multiply.

$$64 \times 100$$

Product: _____

$$3,500 \div 10 = ?$$

- A.** 350
- B.** 3,510
- C.** 35
- D.** 35,000

Divide.

$$9,000 \div 100$$

Quotient: _____

True or false?

$$500 \div 10 = 5$$

A. True

B. False

SET 11 • 5.NBT.A.2

FORWARD 1

Compute fluently with multi-digit decimals • Grade 6 • The Number System •
6.NS.B.3

Multiply.

$$2.4 \times 0.35$$

Product: _____

Divide.

$$1.44 \div 0.12$$

Quotient: _____

Multiply.

$$12.5 \times 0.8$$

A. 100

B. 10

C. 0.1

D. 1

Add.

$$0.875 + 12.4$$

Sum: _____

True or false?

$$4.5 \div 0.05 = 9$$

A. True

B. False

FORWARD 2

Powers of 10 with negative rational numbers • Grade 7 • The Number System •
7.NS.A.2.c

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Find the value.

$$-0.45 \times 100$$

Value: _____

Find the value.

$$-63.5 \div 1,000$$

Value: _____

Find the value.

$$(-2.7) \times (-10)$$

A. -0.27

B. -27

C. 0.27

D. 27

Find the value.

$$0.8 \times (-100)$$

Value: _____

True or false?

$$-5.2 \div 100 = -0.052$$

A. True

B. False

Explain patterns in the number of zeros and the placement of the decimal point

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	A digit is worth 10 times as much as in the place to its right	4.NBT.A.1
BACKWARD 2	Tenths as hundredths	4.NF.C.5
FORWARD 1	Explain decimal placement when multiplying and dividing decima...	6.NS.B.3
FORWARD 2	Write fractions as decimals and explain why they terminate or rep...	7.NS.A.2.d

SET 59 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Explain why $46 \times 1,000 = 46,000$ has three more zeros than 46.

Explanation: _____

Why is $5.7 \times 100 = 570$?

- A.** You write two zeros after 5.7.
- B.** Each digit becomes worth 100 times as much, so it moves 2 places to the left.
- C.** Each digit becomes worth $\frac{1}{100}$ as much.
- D.** You move the decimal point 2 places to the left.

True or false?

$38 \div 10 = 3.8$ because dividing by 10 makes each digit worth $\frac{1}{10}$ as much, so each digit moves one place to the right.

A. True

B. False

Mia says $2.4 \times 10 = 2.40$ because "multiplying by 10 adds a zero."

Give the correct product and explain her mistake.

Correct product: _____

Explanation: _____

$$10^3 = 1,000.$$

Explain how the exponent tells you the number of zeros.

Explanation: _____

SET 59 • 5.NBT.A.2

BACKWARD 1

A digit is worth 10 times as much as in the place to its right • Grade 4 •
Number & Operations in Base Ten • 4.NBT.A.1

True or false?

In 770, the 7 in the hundreds place is worth 10 times the 7 in the tens place.

A. True

B. False

Why is $30 \times 10 = 300$?

- A.** The 3 tens become 3 hundreds.
- B.** The 3 moves one place to the right.
- C.** Each digit becomes worth less.
- D.** You add 10 to 30.

How many tens are equal to 6 hundreds?

Tens: _____

True or false?

In 5,505, the 5 in the thousands place is worth 10 times the 5 in the ones place.

- A. True
- B. False

Write the number that is 10 times as much as 408.

Number: _____

SET 59 • 5.NBT.A.2

BACKWARD 2

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Write $\frac{3}{10}$ as a fraction with denominator 100.

Fraction: _____

True or false?

$$\frac{6}{10} = \frac{60}{100}$$

A. True

B. False

$$\frac{40}{100} = ?$$

A. $\frac{40}{10}$

B. $\frac{400}{10}$

C. $\frac{4}{10}$

D. $\frac{4}{100}$

Add.

$$\frac{2}{10} + \frac{5}{100}$$

Sum: _____

True or false?

$$\frac{7}{10} = \frac{7}{100}$$

A. True

B. False

SET 59 • 5.NBT.A.2

FORWARD 1

Explain decimal placement when multiplying and dividing decimals • Grade 6 •
The Number System • 6.NS.B.3

Multiply.

$$0.3 \times 0.2$$

Product: _____

Why does $0.4 \times 0.03 = 0.012$ have three decimal places?

- A.** Because $4 \times 3 = 12$.
- B.** Tenths times hundredths gives thousandths.
- C.** Because 0.03 is greater than 0.4.
- D.** Because the product is greater than 1.

Divide.

$$7.2 \div 0.08$$

Quotient: _____

True or false?

$$1.25 \times 0.4 = 0.5$$

A. True

B. False

Explain why $4.8 \div 0.6$ has the same quotient as $48 \div 6$.

Quotient: _____

Explanation: _____

FORWARD 2

Write fractions as decimals and explain why they terminate or repeat • Grade 7

• The Number System • 7.NS.A.2.d

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

True or false?

$\frac{3}{40}$ can be written as a decimal that ends.

- A. True
- B. False

Which decimal equals $\frac{7}{25}$?

A. 0.35

B. 0.28

C. 2.8

D. 0.07

Use long division to write $\frac{5}{12}$ as a decimal. Show your work.

Decimal: _____

True or false?

$$\frac{2}{3} = 0.67 \text{ exactly.}$$

A. True

B. False

Read and write decimals to thousandths using numerals and words

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Decimals for tenths and hundredths	4.NF.C.6
BACKWARD 2	Reading and writing whole numbers	4.NBT.A.2
BACKWARD 3	Naming tenths and hundredths	4.NF.C.5
FORWARD 1	Find and position rational numbers on a number line	6.NS.C.6.c
FORWARD 2	Write fractions as decimals	7.NS.A.2.d

SET 12 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write the number in standard form.

four and twenty-six hundredths

Number: _____

Write 3.508 in words.

Words: _____

Which number is nine and forty-one thousandths?

- A.** 9.410
- B.** 9.41
- C.** 9.0041
- D.** 9.041

Write the number in standard form.

seventeen and three tenths

Number: _____

True or false?

0.072 is read "seventy-two hundredths."

A. True

B. False

SET 12 • 5.NBT.A.3.a

BACKWARD 1

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{6}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.5 = \frac{5}{10}$$

A. True

B. False

Which decimal is equal to $\frac{12}{100}$?

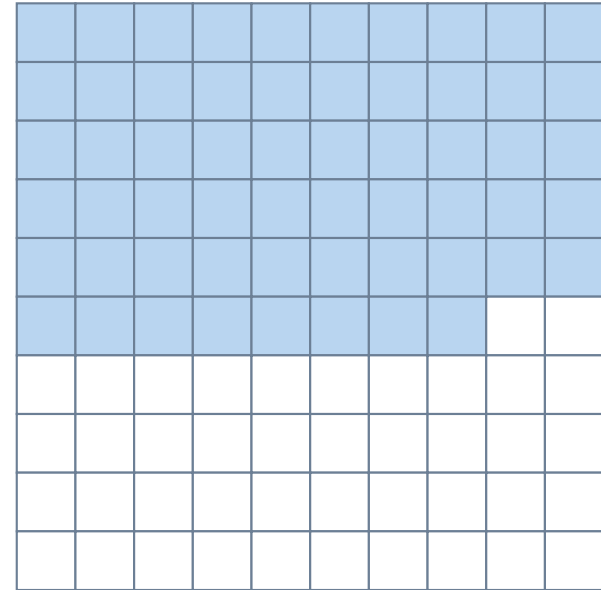
- A.** 1.2
- B.** 12.0
- C.** 0.12
- D.** 0.012

Write $3\frac{7}{10}$ as a decimal.

Decimal: _____

What decimal does the shaded part of the grid show?

Decimal: _____



SET 12 • 5.NBT.A.3.a

BACKWARD 2

Reading and writing whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write the number in standard form.

two thousand, four hundred six

Number: _____

True or false?

5,030 is read "five thousand three hundred."

A. True

B. False

Which number is eighty thousand, nine hundred?

A. 80,900

B. 8,900

C. 89,000

D. 80,090

Write 607 in words.

Words: _____

Write the number in standard form.

twelve thousand, fifty

Number: _____

SET 12 • 5.NBT.A.3.a

BACKWARD 3

Naming tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

True or false?

Three tenths is equal to thirty hundredths.

- A. True
- B. False

How many hundredths are in 4 tenths?

Hundredths: _____

Which fraction is "five hundredths"?

A. $\frac{100}{5}$

B. $\frac{5}{10}$

C. $\frac{50}{100}$

D. $\frac{5}{100}$

Write "nine tenths" as a fraction.

Fraction: _____

True or false?

$$\frac{60}{100} = \frac{6}{10}$$

A. True

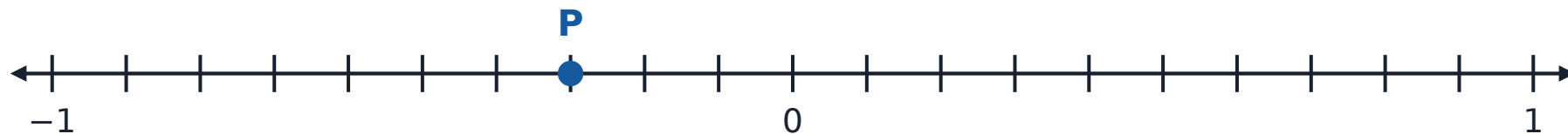
B. False

SET 12 • 5.NBT.A.3.a

FORWARD 1

Find and position rational numbers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?

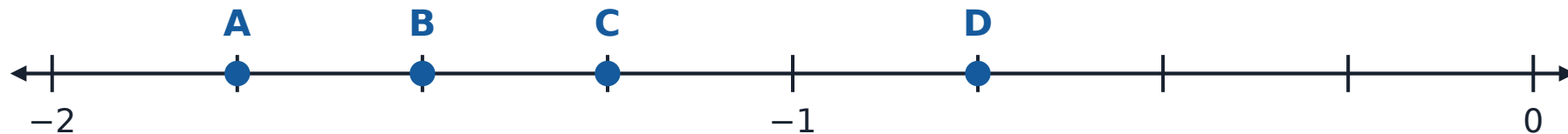


P = _____

Plot and label point Q at 0.45 on the number line.



Which point is at -1.25 ?



A. Point A

B. Point B

C. Point C

D. Point D

What number does point R represent?



R = _____

True or false?

On a number line, -0.6 is to the left of -0.5 .

A. True

B. False

SET 12 • 5.NBT.A.3.a

FORWARD 2

Write fractions as decimals • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

Use long division to write $\frac{5}{12}$ as a decimal. Show your work.

Decimal: _____

$$\frac{11}{20} = ?$$

A. 0.55

B. 5.5

C. 0.2

D. 0.11

Use long division to write $\frac{4}{9}$ as a decimal. Show your work.

Decimal: _____

True or false?

$$\frac{7}{25} = 0.32$$

A. True

B. False

Write decimals to thousandths in expanded form

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Expanded form of whole numbers	4.NBT.A.2
BACKWARD 2	Decimals and fractions with denominators 10 and 100	4.NF.C.6
FORWARD 1	Write expanded form with exponents	6.EE.A.1
FORWARD 2	Decimal expansions of fractions — nearest related (no direct Gra...	7.NS.A.2.d

SET 13 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write 352.68 in expanded form.

Expanded form: _____

Write the number in standard form.

$$4 \times 10 + 7 \times 1 + 3 \times \frac{1}{10} + 9 \times \frac{1}{1000}$$

Number: _____

Which is the expanded form of 6.045?

A. $6 + 45$

B. $6 \times 10 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$

C. $6 \times 1 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$

D. $6 \times 1 + 4 \times \frac{1}{10} + 5 \times \frac{1}{100}$

Write the number in standard form.

$$8 \times 100 + 2 \times \frac{1}{10} + 6 \times \frac{1}{100}$$

Number: _____

True or false?

$$0.908 = 9 \times \frac{1}{10} + 8 \times \frac{1}{1000}$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

BACKWARD 1

Expanded form of whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write 4,371 in expanded form.

Expanded form: _____

True or false?

$$6,052 = 6,000 + 500 + 2$$

A. True

B. False

Which number is $7,000 + 400 + 9$?

- A.** 7,409
- B.** 7,490
- C.** 74,009
- D.** 7,049

Write $3 \times 1,000 + 8 \times 100 + 5 \times 1$ in standard form.

Number: _____

True or false?

$$900 + 20 + 4 = 9,024$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

BACKWARD 2

Decimals and fractions with denominators 10 and 100 • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 0.6 as a fraction.

Fraction: _____

True or false?

$$\frac{4}{100} = 0.4$$

A. True

B. False

Which fraction is equal to 0.25?

A. $\frac{1}{25}$

B. $\frac{25}{100}$

C. $\frac{25}{10}$

D. $\frac{2}{5}$

Write $2 + \frac{3}{10} + \frac{7}{100}$ as a decimal.

Decimal: _____

True or false?

$$\frac{9}{10} = 0.09$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

FORWARD 1

Write expanded form with exponents • Grade 6 • Expressions & Equations •
6.EE.A.1

Write $5,000 + 300 + 20 + 7$ using powers of 10.

Expression: _____

Evaluate.

$$4 \times 10^3 + 6 \times 10^1$$

Value: _____

Which number is equal to 10^4 ?

- A.** 40
- B.** 100,000
- C.** 1,000
- D.** 10,000

Write $600 + 90 + 1$ using powers of 10.

Expression: _____

True or false?

$$2 \times 10^2 + 5 = 250$$

A. True

B. False

FORWARD 2

Decimal expansions of fractions • Grade 7 • The Number System • 7.NS.A.2.d

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Write $\frac{3}{4}$ as a decimal.

Decimal: _____

Use long division to write $\frac{1}{6}$ as a decimal. Show your work.

Decimal: _____

Which fraction is equal to 0.666... (the 6 repeats)?

A. $\frac{3}{5}$

B. $\frac{6}{10}$

C. $\frac{2}{3}$

D. $\frac{1}{6}$

Use long division to write $\frac{5}{8}$ as a decimal. Show your work.

Decimal: _____

True or false?

$$\frac{2}{5} = 0.4$$

A. True

B. False

Compare decimals to thousandths

MAIN	Grade 5 main question	5.NBT.A.3.b
BACKWARD 1	Comparing decimals to hundredths	4.NF.C.7
BACKWARD 2	The symbols $>$, $<$, and $=$	1.NBT.B.3
FORWARD 1	Order rational numbers	6.NS.C.7.b
FORWARD 2	Compare fractions and decimals using decimal expansions	7.NS.A.2.d

SET 14 • 5.NBT.A.3.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Write $>$, $<$, or $=$ to compare.

0.47 0.405

Symbol: _____

Write $>$, $<$, or $=$ to compare.

3.06 3.060

Symbol: _____

Which number is greatest?

- A.** 5.2
- B.** 5.19
- C.** 5.093
- D.** 5.199

Order from least to greatest.

2.31, 2.301, 2.13, 2.3

Order: _____

True or false?

$$0.8 < 0.79$$

A. True

B. False

SET 14 • 5.NBT.A.3.b

BACKWARD 1

Comparing decimals to hundredths • Grade 4 • Number & Operations–Fractions
• 4.NF.C.7

Write $>$, $<$, or $=$ to compare.

0.6 0.58

Symbol: _____

True or false?

$$0.3 = 0.03$$

A. True

B. False

Which number is less?

A. 0.09

B. 0.1

Write $>$, $<$, or $=$ to compare.

0.75 0.8

Symbol: _____

True or false?

$$0.42 > 0.5$$

A. True

B. False

BACKWARD 2

The symbols $>$, $<$, and $=$ • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

True or false?

$$9 > 6$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

52 25

Symbol: _____

True or false?

The symbol $>$ means "is less than."

A. True

B. False

Which statement is true?

A. $14 < 41$

B. $14 = 41$

C. $14 > 41$

SET 14 • 5.NBT.A.3.b

FORWARD 1

Order rational numbers • Grade 6 • The Number System • 6.NS.C.7.b

Write $>$, $<$, or $=$ to compare.

-0.4 ____ -0.45

Symbol: _____

Which number is least?

- A.** -2
- B.** -2.5
- C.** 0.5
- D.** -2.05

Order from least to greatest.

-1.2 , 0.3 , -1.25 , 0

Order: _____

True or false?

$$-3.7 > -3.6$$

A. True

B. False

A temperature of -2.5°C is colder than a temperature of -1.8°C .

Write an inequality to show this.

Inequality: _____

SET 14 • 5.NBT.A.3.b

FORWARD 2

Compare fractions and decimals using decimal expansions • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Is $\frac{5}{8}$ greater than or less than 0.6?

Decimal: _____

Answer: _____

Which number is greatest?

A. $\frac{4}{5}$

B. 0.7

C. $\frac{3}{4}$

D. 0.78

Order from least to greatest.

$$\frac{2}{3}, 0.6, \frac{5}{8}$$

Order: _____

True or false?

$$\frac{1}{3} > 0.33$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

$$\frac{7}{20} \text{ ____ } 0.4$$

Symbol: _____

Round decimals to any place

MAIN

Grade 5 main question

5.NBT.A.4

BACKWARD 1

Rounding whole numbers

4.NBT.A.3

BACKWARD 2

Decimals on a number line

4.NF.C.6

FORWARD 1

Estimate to check computations with decimals

6.NS.B.3

FORWARD 2

Use estimation to judge whether an answer is reasonable

7.EE.B.3

SET 15 • 5.NBT.A.4

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 6.483 to the nearest tenth.

Rounded: _____

Round 12.749 to the nearest hundredth.

Rounded: _____

Round 0.865 to the nearest whole number.

A. 0.9

B. 1

C. 0

D. 0.87

Round 39.96 to the nearest tenth.

Rounded: _____

True or false?

4.351 rounded to the nearest hundredth is 4.36.

A. True

B. False

SET 15 • 5.NBT.A.4

BACKWARD 1

Rounding whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.3

Round 4,682 to the nearest hundred.

Rounded: _____

True or false?

350 rounded to the nearest hundred is 300.

A. True

B. False

Round 7,249 to the nearest thousand.

- A.** 7,200
- B.** 7,000
- C.** 7,250
- D.** 8,000

Round 85 to the nearest ten.

Rounded: _____

True or false?

1,349 rounded to the nearest hundred is 1,400.

- A. True
- B. False

SET 15 • 5.NBT.A.4

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal is at point A?



A = _____

True or false?

0.7 is closer to 1 than to 0.

A. True

B. False

Is 0.43 closer to 0.4 or to 0.5?

A. 0.4

B. 0.5

What decimal is at point B?



B = _____

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

SET 15 • 5.NBT.A.4

FORWARD 1

Estimate to check computations with decimals • Grade 6 • The Number System
• 6.NS.B.3

Round each number to the nearest whole number. Then add to estimate.

$$18.7 + 6.25$$

Estimate: _____

Which is the best estimate of 4.9×6.1 ?

A. 36

B. 30

C. 24

D. 11

Estimate $31.8 \div 7.9$ by rounding each number to the nearest whole number.

Estimate: _____

True or false?

29.6 – 10.2 is about 40.

A. True

B. False

Kim says $3.9 \times 2.1 = 81.9$.

Use an estimate to decide whether her answer is reasonable. Write yes or no.

Answer: _____

SET 15 • 5.NBT.A.4

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A \$24.75 shirt is 20% off.

Round the price to the nearest dollar. Then use the rounded price to estimate the sale price.

Rounded price: _____

Estimated sale price: _____

A worker earns \$19.85 per hour for 9.75 hours.
Which is the best estimate of her pay?

- A.** \$20
- B.** \$2,000
- C.** \$200
- D.** \$100

True or false?

A 6.2% tax on \$49.50 is about \$3.

A. True

B. False

Nina says $\frac{3}{4} \times 39.8$ is about 40.

Round 39.8 to the nearest whole number and estimate $\frac{3}{4} \times 39.8$. Is her estimate reasonable?

Your estimate: _____

Reasonable? _____

A board is $8\frac{1}{4}$ feet long. It is cut into pieces that are $1\frac{7}{8}$ feet long.

Estimate how many whole pieces can be cut.

Estimate: _____

Multiply multi-digit whole numbers using the standard algorithm

MAIN	Grade 5 main question	5.NBT.B.5
BACKWARD 1	Multiplying by a one-digit or two-digit number	4.NBT.B.5
BACKWARD 2	Multiplication facts	3.OA.C.7
BACKWARD 3	Adding partial products	4.NBT.B.4
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply integers	7.NS.A.2.c

SET 16 • 5.NBT.B.5

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$246 \times 37$$

Product: _____

Use the standard algorithm to multiply. Show your work.

$$1,385 \times 24$$

Product: _____

This is the standard algorithm for 508×63 with one partial product missing.
What is the missing partial product?

$$\begin{array}{r} 508 \\ \times 63 \\ \hline 1,524 \\ + \quad ? \\ \hline 32,004 \end{array}$$

A. 1,524

B. 31,904

C. 30,480

D. 3,048

Use the standard algorithm to multiply. Show your work.

$$79 \times 86$$

Product: _____

A theater has 48 rows with 125 seats in each row.

Use the standard algorithm to find how many seats are in the theater. Show your work.

Seats: _____

SET 16 • 5.NBT.B.5

BACKWARD 1

Multiplying by a one-digit or two-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$346 \times 7$$

Product: _____

Multiply.

$$23 \times 45$$

Product: _____

True or false?

$$1,203 \times 4 = 4,082$$

A. True

B. False

$$62 \times 18 = ?$$

A. 1,126

B. 496

C. 1,116

D. 1,006

Multiply.

$$58 \times 30$$

Product: _____

SET 16 • 5.NBT.B.5

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

Multiply.

$$8 \times 9$$

Product: _____

True or false?

$$6 \times 9 = 56$$

A. True

B. False

$$4 \times 8 = ?$$

A. 24

B. 36

C. 32

D. 28

Multiply.

$$9 \times 9$$

Product: _____

SET 16 • 5.NBT.B.5

BACKWARD 3

Adding partial products • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Add.

$$7,380 + 1,722$$

Sum: _____

Add.

$$27,700 + 5,540$$

Sum: _____

True or false?

$$30,480 + 1,524 = 31,904$$

A. True

B. False

$$6,400 + 394 = ?$$

A. 6,784

B. 6,794

C. 7,794

D. 6,694

Add.

$$4,800 + 1,200$$

Sum: _____

SET 16 • 5.NBT.B.5

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$2.46 \times 3.7$$

Product: _____

Multiply.

$$13.85 \times 2.4$$

Product: _____

Multiply.

$$5.08 \times 6.3$$

A. 31.904

B. 320.04

C. 3.2004

D. 32.004

Multiply.

$$0.79 \times 8.6$$

Product: _____

True or false?

$$1.25 \times 4.8 = 60$$

A. True

B. False

SET 16 • 5.NBT.B.5

FORWARD 2

Multiply integers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-24) \times 15$$

Product: _____

Find the product.

$$(-12) \times (-35)$$

Product: _____

Find the product.

$$18 \times (-21)$$

- A.** 378
- B.** -378
- C.** -39
- D.** -388

Find the product.

$$(-3) \times (-4) \times (-25)$$

Product: _____

True or false?

$$(-6)(-7) = -42$$

A. True

B. False

Divide with up to four-digit dividends and two-digit divisors

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Multiplying to check a quotient	4.NBT.B.5
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Divide multi-digit numbers using the standard algorithm	6.NS.B.2
FORWARD 2	Divide integers	7.NS.A.2.b

SET 17 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$2,016 \div 24$$

Quotient: _____

Divide.

$$7,830 \div 45$$

Quotient: _____

Divide.

$$952 \div 17$$

A. 54

B. 46

C. 56

D. 57

Divide.

$$3,724 \div 38$$

Quotient: _____

Divide. Write the quotient and the remainder.

$$1,000 \div 32$$

Quotient: _____

Remainder: _____

SET 17 • 5.NBT.B.6

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 6$$

Quotient: _____

Divide.

$$1,935 \div 5$$

Quotient: _____

True or false?

$$728 \div 8 = 901$$

A. True

B. False

$$504 \div 7 = ?$$

A. 82

B. 71

C. 62

D. 72

Divide. Write the quotient and the remainder.

$$95 \div 4$$

Quotient and remainder: _____

SET 17 • 5.NBT.B.6

BACKWARD 2

Multiplying to check a quotient • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

True or false?

To check $2,016 \div 24 = 84$, you can multiply 24×84 .

- A. True
- B. False

Multiply to check $2,016 \div 24 = 84$.

$$24 \times 84$$

Product: _____

Which product checks $952 \div 17 = 56$?

- A.** $17 + 56$
- B.** 17×56
- C.** $56 - 17$
- D.** 952×17

Multiply.

$$38 \times 98$$

Product: _____

True or false?

$$32 \times 31 + 8 = 1,000$$

A. True

B. False

SET 17 • 5.NBT.B.6

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$72 \div 8$$

Quotient: _____

Divide.

$$54 \div 9$$

Quotient: _____

True or false?

$$63 \div 7 = 8$$

A. True

B. False

$$40 \div 5 = ?$$

A. 6

B. 7

C. 8

D. 9

Divide.

$$35 \div 7$$

Quotient: _____

SET 17 • 5.NBT.B.6

FORWARD 1

Divide multi-digit numbers using the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$6,432 \div 48$$

Quotient: _____

Use the standard algorithm to divide. Show your work.

$$25,875 \div 69$$

Quotient: _____

Divide.

$$14,040 \div 52$$

- A.** 207
- B.** 2,700
- C.** 270
- D.** 27

Use the standard algorithm to divide. Show your work.

$$9,594 \div 41$$

Quotient: _____

True or false?

$$11,700 \div 45 = 2,600$$

A. True

B. False

SET 17 • 5.NBT.B.6

FORWARD 2

Divide integers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-2,016 \div 24$$

Quotient: _____

Find the quotient.

$$(-952) \div (-17)$$

Quotient: _____

Find the quotient.

$$780 \div (-15)$$

- A.** -52
- B.** -53
- C.** -765
- D.** 52

Find the quotient.

$$-(144 \div 12)$$

Quotient: _____

True or false?

$$(-90) \div 18 = 90 \div (-18)$$

A. True

B. False

Explain division using area models and equations

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Area models for dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Area as length times width	3.MD.C.7.b
BACKWARD 3	Splitting a rectangle to find its area	3.MD.C.7.c
FORWARD 1	Use the distributive property with the greatest common factor	6.NS.B.4
FORWARD 2	Factor and expand linear expressions	7.EE.A.1

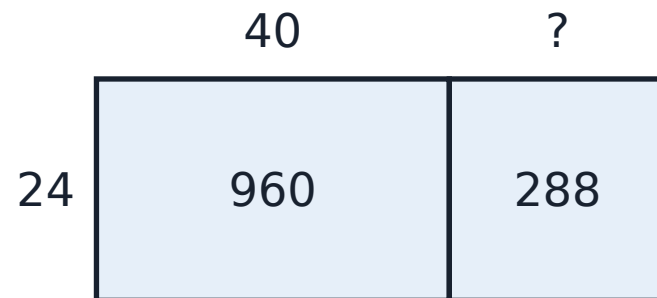
SET 18 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

The area model shows
 $1,248 \div 24$.

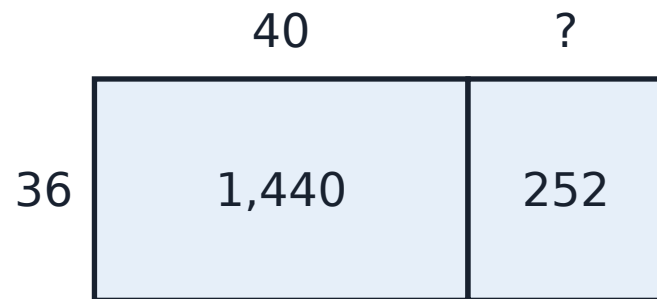
What is the quotient?



Quotient: _____

The area model shows
 $1,692 \div 36$.

What is the quotient?



Quotient: _____

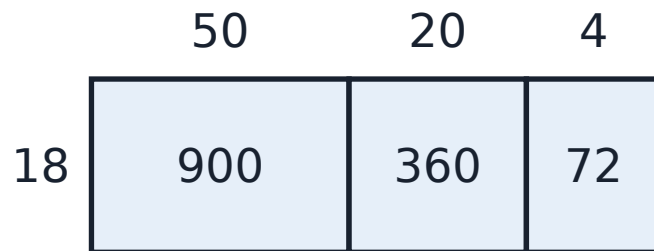
Which equation matches the area model?

A. $1,332 \div 74 = 20$

B. $1,332 \div 18 = 74$

C. $18 \times 50 = 1,332$

D. $900 \div 18 = 74$



Use an area model or equations to divide. Show your model or equations.

$$2,184 \div 28$$

Quotient: _____

True or false?

$$1,248 \div 24 = (960 \div 24) + (288 \div 24)$$

A. True

B. False

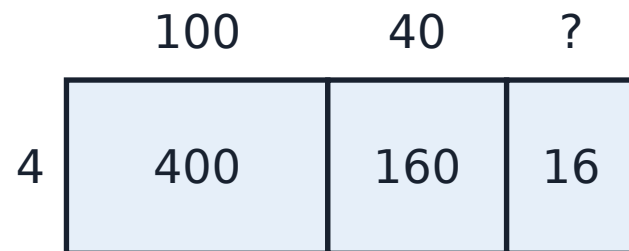
SET 18 • 5.NBT.B.6

BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows
 $576 \div 4$.

What is the quotient?



Quotient: _____

True or false?

$$468 \div 3 = (300 \div 3) + (150 \div 3) + (18 \div 3)$$

A. True

B. False

Which shows $245 \div 5$ split into easier parts?

- A.** $(200 \div 5) + (45 \div 5)$
- B.** $(245 \div 2) + (245 \div 3)$
- C.** $200 \div (5 + 45)$
- D.** $(200 \div 5) \times (45 \div 5)$

Divide.

$$735 \div 7$$

Quotient: _____

A rectangle has an area of 312 square units and a width of 6 units.

What is its length?

Length: _____

SET 18 • 5.NBT.B.6

BACKWARD 2

Area as length times width • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 9 units long and 7 units wide.
What is its area?

Area: _____

True or false?

A rectangle with an area of 40 square units and a side of 5 units has another side of 8 units.

- A. True
- B. False

Which equation gives the area of a 6-by-8 rectangle?

A. $8 - 6 = 2$

B. $6 + 8 = 14$

C. $6 \times 8 = 48$

D. $2 \times (6 + 8) = 28$

A rectangle has an area of 36 square units. One side is 4 units long.

How long is the other side?

Length: _____

Find the area of the rectangle.

Area: _____



SET 18 • 5.NBT.B.6

BACKWARD 3

Splitting a rectangle to find its area • Grade 3 • Measurement & Data •
3.MD.C.7.c

True or false?

The area of a rectangle that is 4 by $(10 + 3)$ equals $4 \times 10 + 4 \times 3$.

A. True

B. False

A 5-by-12 rectangle is split into a 5-by-10 rectangle and a 5-by-2 rectangle.

What is the total area?

Area: _____

A 7-by-13 rectangle is split into a 7-by-10 part and a 7-by-3 part.

Which expression gives its area?

A. $7 \times 10 + 7 \times 3$

B. $7 \times 10 + 3$

C. $7 + 10 + 3$

D. $7 \times 10 \times 3$

Fill in the blank.

$$6 \times 14 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

True or false?

$$8 \times 12 = 8 \times 10 + 2$$

A. True

B. False

SET 18 • 5.NBT.B.6

FORWARD 1

Use the distributive property with the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $36 + 48$ as a product.

Expression: _____

Find the greatest common factor of 32 and 56.

GCF: _____

Which expression is equal to $45 + 60$ and uses the greatest common factor?

- A.** $15(3 + 6)$
- B.** $15(3 + 4)$
- C.** $3(15 + 20)$
- D.** $5(9 + 12)$

Complete the equation.

$$24 + 40 = 8(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

True or false?

$$18 + 27 = 9(2 + 3)$$

A. True

B. False

SET 18 • 5.NBT.B.6

FORWARD 2

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor using the greatest common factor.

$$12x + 30$$

Expression: _____

Expand.

$$7(4y - 3)$$

Expression: _____

Which expression is equivalent to $18a - 24$?

- A.** $3(6a - 4)$
- B.** $6(3a - 24)$
- C.** $18(a - 6)$
- D.** $6(3a - 4)$

Factor out -4 .

$$-8n - 20$$

Expression: _____

True or false?

$$4(2.5x + 3) = 10x + 3$$

A. True

B. False

Draw division models and explain the strategy

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Area models for dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
FORWARD 1	Connect the standard division algorithm to partial quotients	6.NS.B.2
FORWARD 2	Solve real-world problems by dividing rational numbers	7.NS.A.3

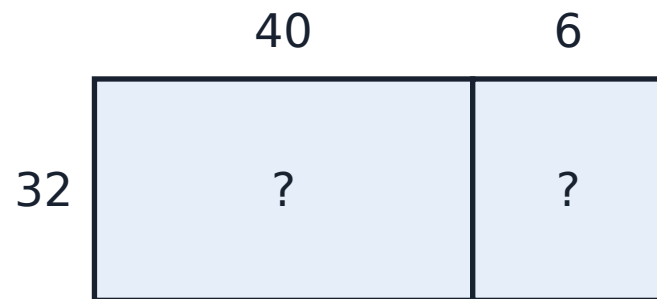
SET 60 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Complete the area model for $1,472 \div 32$ by writing the area of each part.

Then write the quotient.



Quotient: _____

Draw an area model to divide $1,615 \div 19$.

Explain how your model shows the quotient.

Quotient: _____

Explanation: _____

Lena finds $1,380 \div 23$ by thinking $23 \times 60 = 1,380$.

Which statement explains her strategy?

- A.** She multiplied 1,380 by 23 to get 60.
- B.** She divided 23 by 1,380 to get 60.
- C.** Division finds the missing factor:
 $23 \times ? = 1,380$, so the quotient is 60.
- D.** She subtracted 23 from 1,380 to get 60.

Explain how these equations show $2,345 \div 35$.

$$35 \times 60 = 2,100$$

$$35 \times 7 = 245$$

$$2,100 + 245 = 2,345$$

Quotient: _____

Explanation: _____

True or false?

An area model for $1,512 \div 36$ can use parts with areas 1,440 and 72, so the quotient is $40 + 2 = 42$.

- A.** True
- B.** False

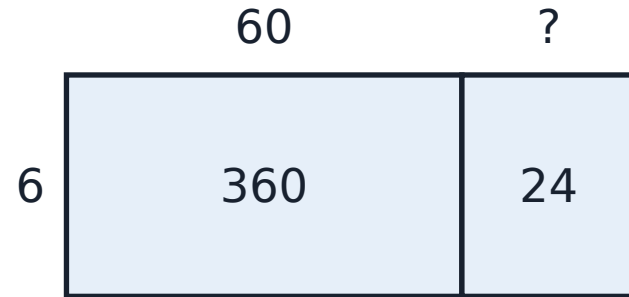
SET 60 • 5.NBT.B.6

BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows
 $384 \div 6$.

What is the missing length?



Length: _____

True or false?

An area model for $528 \div 4$ can be split into $400 \div 4$, $120 \div 4$, and $8 \div 4$.

- A. True
- B. False

An area model for $945 \div 9$ has parts with areas 900 and 45.

What are the lengths of the parts?

A. 100 and 5

B. 90 and 5

C. 10 and 5

D. 100 and 45

Divide.

$$672 \div 8$$

Quotient: _____

True or false?

In an area model for $945 \div 9$, the side length 9 is the quotient.

- A. True
- B. False

SET 60 • 5.NBT.B.6

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find the unknown factor.

$$6 \times ? = 42$$

Factor: _____

True or false?

$32 \div 4 = 8$ because $4 \times 8 = 32$.

A. True

B. False

Which multiplication fact helps you find $56 \div 7$?

A. $8 + 7 = 15$

B. $7 \times 7 = 49$

C. $56 \times 7 = 392$

D. $7 \times 8 = 56$

Find the unknown factor.

$$? \times 9 = 81$$

Factor: _____

True or false?

To find $27 \div 3$, you can ask "27 times what number is 3?"

- A. True
- B. False

SET 60 • 5.NBT.B.6

FORWARD 1

Connect the standard division algorithm to partial quotients • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$8,640 \div 36$$

Quotient: _____

In the standard algorithm for $5,712 \div 28$, the first step uses $28 \times 200 = 5,600$.

What does the 200 stand for?

- A.** Part of the quotient
- B.** The dividend
- C.** The remainder
- D.** Part of the divisor

True or false?

The steps of the standard algorithm for $1,836 \div 17$ match an area model with parts 17×100 and 17×8 .

A. True

B. False

Use the standard algorithm to divide. Show your work.

$$13,248 \div 46$$

Quotient: _____

True or false?

$$3,825 \div 45 = 95$$

A. True

B. False

SET 60 • 5.NBT.B.6

FORWARD 2

Solve real-world problems by dividing rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver goes down 126 feet in 18 equal stages.

Write the change in depth for each stage as a signed number.

Change: _____

A business lost \$4,320 over 24 months, the same amount each month.

Which number shows the change each month?

A. $-\$108$

B. $\$180$

C. $-\$1,800$

D. $-\$180$

The temperature fell 22.5°F in 9 hours at a steady rate.
Write the change per hour as a signed number.

Change: _____

True or false?

A change of $-\$1,512$ shared equally over 36 weeks is a change of $-\$42$ each week.

A. True

B. False

A tank loses $\frac{3}{4}$ gallon every hour.

After how many hours is the change in the amount of water -6 gallons?

Hours: _____

Add decimals to hundredths

MAIN

Grade 5 main question

5.NBT.B.7

BACKWARD 1 Adding multi-digit whole numbers

4.NBT.B.4

BACKWARD 2 Adding tenths and hundredths

4.NF.C.5

BACKWARD 3 Writing decimals with the same number of places

4.NF.C.6

FORWARD 1 Add multi-digit decimals

6.NS.B.3

FORWARD 2 Add rational numbers

7.NS.A.1.d

SET 19 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$23.45 + 8.7$$

Sum: _____

Add.

$$15.3 + 6.48$$

Sum: _____

Add.

$$4.06 + 12.9$$

- A.** 16.15
- B.** 16.069
- C.** 5.35
- D.** 16.96

Add.

$$9.65 + 0.8$$

Sum: _____

Maya had \$18.50. She earned \$6.75 more.
How much money does she have now?

Money: _____

SET 19 • 5.NBT.B.7

BACKWARD 1

Adding multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Add.

$$2,345 + 870$$

Sum: _____

Add.

$$1,530 + 648$$

Sum: _____

True or false?

$$406 + 1,290 = 1,596$$

A. True

B. False

$$1,265 + 735 = ?$$

- A.** 2,100
- B.** 2,000
- C.** 1,900
- D.** 1,990

Add.

$$1,850 + 675$$

Sum: _____

SET 19 • 5.NBT.B.7

BACKWARD 2

Adding tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

Add.

$$\frac{3}{10} + \frac{45}{100}$$

Sum: _____

True or false?

$$\frac{6}{10} + \frac{2}{100} = \frac{8}{100}$$

A. True

B. False

$$\frac{7}{10} + \frac{15}{100} = ?$$

A. $\frac{85}{100}$

B. $\frac{22}{100}$

C. $\frac{22}{110}$

D. $\frac{85}{10}$

Write $\frac{4}{10}$ as hundredths.

Fraction: _____

True or false?

$$\frac{5}{10} + \frac{5}{100} = \frac{10}{100}$$

A. True

B. False

SET 19 • 5.NBT.B.7

BACKWARD 3

Writing decimals with the same number of places • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

True or false?

$$8.7 = 8.70$$

A. True

B. False

Write 15.3 with two decimal places.

Decimal: _____

Which decimal is equal to 20?

- A.** 20.00
- B.** 0.20
- C.** 2.00
- D.** 200.0

True or false?

$$6.4 = 6.04$$

A. True

B. False

Write 12.9 with a digit in the hundredths place, without changing its value.

Decimal: _____

SET 19 • 5.NBT.B.7

FORWARD 1

Add multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

$$128.47 + 9.605$$

Sum: _____

Add.

$$54.2 + 18.376$$

Sum: _____

Add.

$$0.952 + 3.48 + 11.6$$

A. 16.132

B. 4.53

C. 15.032

D. 16.032

Add.

$$99.6 + 0.485$$

Sum: _____

True or false?

$$7.3 + 2.145 = 2.875$$

A. True

B. False

SET 19 • 5.NBT.B.7

FORWARD 2

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-12.5 + 4.75$$

Value: _____

Find the value.

$$3.2 + (-8.45)$$

Value: _____

Find the value.

$$-6.3 + (-2.85)$$

- A.** 9.15
- B.** -9.15
- C.** -3.45
- D.** 3.45

Find the value.

$$-4.6 + 10.25$$

Value: _____

True or false?

$$-1.5 + 1.5 = 0$$

A. True

B. False

Subtract decimals to hundredths

MAIN

Grade 5 main question

5.NBT.B.7

BACKWARD 1

Subtracting multi-digit whole numbers

4.NBT.B.4

BACKWARD 2

Subtracting mixed numbers in tenths and hundredths

4.NF.B.3.c

FORWARD 1

Subtract multi-digit decimals

6.NS.B.3

FORWARD 2

Subtract rational numbers

7.NS.A.1.d

SET 61 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Subtract.

$$15.3 - 6.48$$

Difference: _____

Subtract.

$$20 - 7.35$$

Difference: _____

Maya had \$18.50. She spent \$6.75.

How much money does she have left?

Money left: _____

Subtract.

$$9.04 - 3.7$$

A. 6.34

B. 5.34

C. 8.67

D. 5.44

True or false?

$$6.5 - 2.75 = 3.75$$

A. True

B. False

SET 61 • 5.NBT.B.7

BACKWARD 1

Subtracting multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Subtract.

$$1,530 - 648$$

Difference: _____

$$2,000 - 735 = ?$$

A. 2,735

B. 1,275

C. 1,265

D. 1,365

Subtract.

$$1,850 - 675$$

Difference: _____

True or false?

$$5,000 - 1,234 = 3,766$$

A. True

B. False

Subtract.

$$4,206 - 1,519$$

Difference: _____

SET 61 • 5.NBT.B.7

BACKWARD 2

Subtracting mixed numbers in tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Subtract.

$$5\frac{3}{10} - 2\frac{6}{10}$$

Difference: _____

True or false?

$$4 - \frac{35}{100} = 3\frac{65}{100}$$

A. True

B. False

$$6\frac{2}{10} - 1\frac{9}{10} = ?$$

A. $5\frac{7}{10}$

B. $4\frac{7}{10}$

C. $4\frac{3}{10}$

D. $5\frac{3}{10}$

Subtract.

$$3\frac{40}{100} - 1\frac{25}{100}$$

Difference: _____

True or false?

$$2\frac{1}{10} - \frac{5}{10} = 2\frac{4}{10}$$

A. True

B. False

SET 61 • 5.NBT.B.7

FORWARD 1

Subtract multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Subtract.

$$54.2 - 18.376$$

Difference: _____

Subtract.

$$100 - 0.485$$

Difference: _____

True or false?

$$7.3 - 2.145 = 5.245$$

A. True

B. False

Subtract.

$$12.06 - 4.875$$

Difference: _____

Subtract.

$$30 - 0.009$$

A. 29.991

B. 30.009

C. 29.001

D. 29.91

SET 61 • 5.NBT.B.7

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$3.2 - 8.45$$

Value: _____

Find the value.

$$-4.6 - (-10.25)$$

Value: _____

Find the value.

$$-2.75 - 4.5$$

Value: _____

Find the value.

$$1.2 - (-3.65)$$

- A.** 2.45
- B.** -4.85
- C.** -2.45
- D.** 4.85

True or false?

$$-8.1 - (-8.1) = 0$$

A. True

B. False

Use place-value models to add and subtract decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Add tenths and hundredths	4.NF.C.5
BACKWARD 2	Rename a unit as ten of the next smaller unit	4.NBT.A.1
FORWARD 1	Add and subtract multi-digit decimals with the standard algorithm	6.NS.B.3
FORWARD 2	Add and subtract rational numbers	7.NS.A.1.d

SET 62 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Shade the grid to show
 $0.4 + 0.25$.

Then write the sum.

Sum: _____

